

ERP All-in-One Document

Welcome.

This all-in-one document includes all information for teachers, students, and teaching assistants (TAs) for experimental research project, 2026. Please send an email to Hai Wang (h.wang5@uu.nl) and cc Marijn van Huis (M.A.vanHuis@uu.nl) if something is unclear or missing. We will keep the documents updated while do our best to keep the structure logical and clear.

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1. Introduction to the course and schedule

The aim of the course is to let students go through a complete experimental cycle from (1) designing their own research question, (2) developing an experimental setup/protocol that can answer their research question, (3) performing data analysis and necessary simulation, (4) communicating the results to the public, and (5) reflecting on own study and performance.

Students are the owner of the project and should enjoy the pleasure of finding things out!

Schedule

The course will last for 10 weeks and consists of 3 project phases (see the details in the attached schedule).

In **phase 1** (first 2 weeks 22/04 to 01/05, 2026), we will provide students with necessary modules and basic experimental knowledge/technical/soft skills that you need for your projects.

-In week 1, students will follow MANDATORY introduction lectures related to: the projects (9-12:30 am, 22/04), the course overview including the schedule, implementation and practical suggestion of poster design for the scientific communication + question session (13:15-15:00, 22/04), how to work in a team (9-10:45 am, 24/04)

-In week 2, we provide you 6 modules to students who will be assigned to different modules based on the project that students will be working on (the distribution will be based on suggestion/feedbacks by the teacher who supervises the project; we will randomly distribute three students to three modules suggested by the teacher)

In **phase 2** (week 3-9, 04/05 to 19/06/2026), students will work with their own group on the project that you chose, in the Minnaert building or experimental research groups where your research will be conducted.

In **phase 3** (week 10, on 24/06/2026), students will disseminate their research results in a poster session to their peers and teachers. In a final session, students, TA and teachers will reflect on the students' work to evaluate what they have learned and what could have been improved (ideally after the poster session in week 10).

When	Where	What/Who
Week 0	Online; A link will be provided	Assignment for students: Please form a group of 3 students yourself latest at 13:00, on 21/04. We will help you form a group whenever necessary.
Week 1: 22.04; 09:00-12:00	Minnaert. 001, 002, 003, 004	Lecture 1: project information and project matching Coordinators will provide an overview of the projects that students will work on; then students will break out in 3 lecture

		room, and have the chance to talk to teacher representatives (at least one per institute) who can tell students more details on the project
Week 1: 22-26/04	Online	Assignment for students: Submitting your 5 wishes for the projects that your group would like to work on; a link will be provided on 22/04 after lecture 1. You will be randomly assigned to a project if you fail to submit your choices in time
Week 1: 22.04; 13:15-15:00	KBG Atlas	Lecture 2: Introduction to the course (13:15-15:00; 22.04) Coordinators will present the course structure, schedule, expectation, and implementation to students; Students will get the chance to ask questions to the course coordinators.
Week 1: 24.04; 09-10:45	KBG Atlas	Lecture 3 : Collaboration and team work (09:00-10:45; 24.04) Dr. Ria Dolfine will discuss and reflect together how to work together as a group. The assignment with the deadline on 15 th of May is related to this lecture -so it is important that you and your teammates can join it together.
Week 2: 28.04	Check your Brightspace	Check your Brightspace: (1) Coordinators will release of group-project assignment , and individual student assignment to the modules starting on 29 th /04 (2) Student-module matching: For students: you don't need to choose modules; you will be assigned to one of them, based on the nature of your projects and the suggestion of the teachers -check your Brightspace on 27/04
Week 2: 29.04-01.05 (slot A and D)	M1: Electronics and signal processing (MIN 002) M2: Arduino (MIN001) M3: Laser and optics (MIN 011 and MIN003) M4: Statistical Data Analysis (BBG 079 and BBG017) M5: Mass spectroscopy (MIN011, BBG 169)	Module phase: Students working on one of the 6 modules Time and location: M1: Electronics; Lecture: None, preparation at home (see instruction manual on Brightspace) Project assignment: 29/04: 13.00-17:00 (supervised), MIN002

	M6: Proto lab (Proto lab; CBG 0.22)	01/05: 09.00-13:00 (supervised) and 13.00-17.00 (unsupervised), MIN 002 Supervisor: Marijn van Huis Module 2. ARDUINO Lecture: None. It is efficient to look through the tutorial (see Brightspace) before the first contact hours Project assignment: 29/04: 13.15-17:00 (supervised), MIN0.01 01/05: 09.00-12:45 (unsupervised) and 13.15-17.00 (supervised), MIN0.01 Supervisor: Peter van Capel and Giogio Cover M3: Laser and optics Lecture: 9:30-12:30, 29/04, MIN 011; Group assignment: 2 students/group; 12 groups to build 3 different setups and do measurements 29/04: 13-17:00 (unsupervised), MIN 003, 012; 01/05: 09-17:00 (supervised), MIN 003, 012. Supervisor: Hai Wang Module 4. Statistical data analysis Lecture and individual assignment I: 13:15-17:00, 29/04, BBG 079. Lecture and individual assignment II: 09:00-12:0, 01/05, BBG 017. Supervisor: Anuradha Samajdar Module 5. Mass spectrometry Lecture and individual assignment I: 13:15-17:00, 29/04, MIN 0.11. 09:00-12:0, 01/05, BBG 169. Supervisor: Alexis Gilbert Module 6. Proto Lab Lecture and practice time: 9-12:45, 29/04, CBG 0.22; 9-12:45, 01/05, CBG 0.22 Project assignment: In between lectures, at home Supervisor: Pieter Kooijman
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Week 2: 29/04-01/05	Contact your project teacher	Please contact your teacher by an email to arrange the first meeting in week 3.
Week 3-9:	Minnaert 001, 002, 004, 006 and 012, and other specific labs in other buildings provided by the teacher	Project phase: students will design, construct a setup under the supervision of 1 teacher and 1 TA, supported by technical staff. TAs will be the first contact person for lab issues.
15/05	Uploading to Brightspace	Group assignment: submitting a brief, max 3 page proposal including: (1) the aim of the projects; what is the question to be addressed? (2) your proposal how to solve the problem; and (3) how you will distribute your task, and how you will work together as a team? You can include teacher's and TA's suggestion and feedback on these aspects
Week 10: 22.06, at noon 12:00	Online poster submission to Brightspace	Group assignment: Students will submit their poster to Brightspace (see the suggested format). The coordinator will take care of printing out the posters submitted in time.
Week 10: 24.06 13-16:00	Poster session in MIN Learning Plaza	Students will present their posters
Week 10: 25-27/06	Project reflection and grading of project	Student/teacher/TA will meet to reflect together what went well and wrong. Teacher, with the assistant of TAs will fill out the grading form and send it to Hai Wang . Students will be graded only pass or fail ; we will provide evaluation rubric for PIs to provide feedback to students.

2. Learning goals and objects

Modules

The student follows and completes two of the modules on experimental techniques and methods.

- follows and completes two of the modules on: Electronics and signal processing, Laser and optics; Data processing; Mass spectroscopy; Proto lab; Arduino.
- familiarizes herself/himself with the instrumentation/method used in the modules
- understands the functionality and caveats of the methods and techniques treated in the modules (module-dependent)

Research Project

1. The student defines and formulates an experimental project either entirely by themselves (with teacher and TA's feedback) or from a teacher;
 - formulates an empirically verifiable research question or goal that can be answered or met within the boundary conditions of the project (available time, budget, equipment, expertise)
 - designs or optimize an instrument or setup for an experimental research project given the conditions of the project.
 - searches and reads literature during the definition of the research question or goal and during the design of the experiment.
2. The student plans and executes an experimental project in collaboration.
 - plans and controls a complex experimental project from formulation through construction, test, and execution all the way to completion and reporting.
 - learns to deal with the group dynamics that comes with an (experimental) research project (e.g. division of tasks and responsibilities, coordination, exchange of ideas and information, accountability)
3. The student masters instrumentation
 - chooses the proper instruments and procedures to achieve the desired accuracy and precision.
 - calibrates instruments throughout the project.

understands the physical working principle of the (commercial) instruments, sources and / or detectors used and the possible systematic effects that these introduce.
4. The student is able to construct an experimental setup.
 - builds hand-on experience on building setups and using measurement instruments.
 - builds experience on resolving issues with the setup.

is aware that a constructed setup has its limitations and possible systematic errors.
5. The student takes part in peer consultation and discussion.
 - studies the work of peers substantively in different phases of the project and delivers constructive critique.
 - discusses the research project with peers and can decide which suggestions are improvements and which are not.
6. The student demonstrates conceptual understanding of measured data.
 - is capable of developing a theoretical / computational model for the case that is studied experimentally.
 - can estimate prior to execution of an accurate experiment what the outcomes will be, through calculations and rough estimates given the input values for certain physical quantities.
 - works synthetically and applies (theoretical, technical, or conceptual) knowledge from modules and physics in general in the design of experiments and in the analysis and interpretation of the results.

- formulates scientific arguments following from the research question or goal, the collected and analyzed evidence and conclusions based on results.

Reporting and communication

- 1.** The student communicates and reports effectively, respectfully, convincingly, and scientifically sound.
 - convincingly discusses conclusions and argumentations (founded on theoretical background, experimental research, and logical thinking) orally with the supervisor.
 - has authentic research experience and experiences methods and activities common in research groups.
 - communicate effectively and respectfully with (technical) staff.reports in a scientifically sound way: does not communicate misleading results and does not commit plagiarism.
- 2.** The student documents the project in collaboration at Bachelor Research project level.
 - Documentation includes at the least a (collective) lab journal and a file structure in which all the collected information and data is organized clearly, and furthermore e.g. notes, memos, reports, and presentations for the project group
- 3.** The student creates and presents a poster.
 - can present a poster to fellow researchers.
 - uses proper technical terminology, results including uncertainties, tables and figures and schematics of the setup to make research accessible to peers.
- 4.** The student reflects on own work and achievements during the project and afterwards.
 - monitors the quality of results and improves the design of the experiment during the project.
 - judges and improves own work based on the supplied rubrics and demands.identifies possible, interesting, and relevant follow-up research that follows logically from the conducted research (possibly requiring new equipment or other conditions).

3. Implementation: When/where/who to do what -guides for students/teacher/TAs

To implement these plans, we provide the following guides for students/teachers/TAs for when and what are expected to do from week 1 to 10.

3.1 Guides to students:

Week 0:

Group formation. Please form a group of 3 students yourself **before the first lecture latest at 13:00, on 21/04**. You can choose your own teammates. If you fail to do that in time, we will be in a group of three assigned by a program randomly.

Week 1: 3 Lectures + Project Choosing

Lecture 1: Project information session - in the morning of **22/04**, from **9 to 12:30 am (room 001-004 in Minnaert)**. We will provide you basic information on project topics; at least one teacher from each institute (Debye (including Nanophotonics and Soft Condense Matter and Biophysics (SCMB)), IMAU, GRASP and JULIUS) will be there to provide detailed information in each project from their own institute following a general introduction session by **Hai Wang**. You will have chance to ask the information to the teacher on projects in the break-out rooms.

Lecture 2: Introduction to the course (13:15-15:00; 22.04; KBG Atlas, Lecturer: Hai Wang and Marijn van Huis)

Coordinators will present the course structure, schedule, expectation, and implementation to students; Students will get the chance to ask questions to the course coordinators.

Lecture 3: Collaboration and team work (09:00-10:45; 24.04; KBG Atlas Lecturer: Ria Dolfing): Dr. Ria Dolfing will discuss and reflect together with students how to work together as a group

Project choosing and assignments: Once the group is formed, and after the project information session, students should meet to discuss their choice of projects.

There are **two possibilities** here: **First**, you may propose a project yourself and **discuss its feasibility with Hai Wang and Marijn van Huis**. In the 2024-2025 ERP course, two pilot groups have taken this route. They reported that they learned a lot but also faced significant challenges. Open projects will be evaluated slightly differently, with more weight on the design and planning phase. This means that you can pass the course even if the final target is not fully achieved, provided you can discuss the results and justify how the project could be improved in the future. Because such projects require substantial effort and technical support from technicians, we will limit the number of open projects to **five**.

Second, we will provide a list of possible topics proposed by the teachers (see the project list and descriptions at the end of this document; please also attend the project introduction session on 22/02; 09-12:30 in MIN). You have five days to submit your 5 wishes. You should submit your top five preferred projects online **by 26/04**; a link for it will be provided later.

We will then distribute students to PIs as evenly and fairly as possible, using a code developed by one of our colleagues to minimize “unhappiness.”

Based on experience from last year, some “hot” topics attracted many students (up to 15 times for the most popular one!), which resulted in ~30% of groups not getting any of their top choices. However, **the survey showed that all students still enjoyed their projects even if they didn’t get their first choices. So don’t worry if you don’t get your dream project!**

Hai Wang will be the contact person for group formation and project distribution and provide you help whenever needed.

Week 2: Modules

Based on the project that you chose or are assigned to, you will join one of the modules in week 2 including: electronics, Arduino, Optics, Mass spectrometry, Data analysis and processing, Protolab. **You don’t need to choose the module yourself**: for each project, the teacher responsible has indicated how three students are distributed into (ideally different) modules. **We will randomly distribute your group members into 2-3 modules relevant to your projects.** As such, we hope to prepare you better for your project phase with basic technical skill or safety information. We will send your Module assignment on **28/04** (27th/04 is a public holiday) on Brightspace, so that you know where to go on **29/04**.

The time slot for the modules are **A and D slots (9:00-17:00 for 30/04, and 09:00-17:00 for 01/05)**. There will be assignments (of different natures from different modules); **it is compulsory that you pass the module assigned to you to pass the course.**

Week 2: 29.04-01.05 (slot A and D)	M1: Electronics and signal processing (MIN 002) M2: Arduino (MIN001) M3: Laser and optics (MIN 011 and MIN003) M4: Statistical Data Analysis (BBG 079 and BBG017) M5: Mass spectroscopy (MIN011, BBG 169) M6: Proto lab (Proto lab; CBG 0.22)	Module phase: Students working on one of the 6 modules Time and location: M1: Electronics; Lecture time: None, preparation at home (see instruction manual on Brightspace) Project assignment: 29/04: 13.00-17:00 (supervised), MIN002 01/05: 09.00-13:00 (supervised) and 13.00-17.00 (unsupervised), MIN 002 Supervisor: Marijn van Huis Module 2. ARDUINO
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		Lecture time: None. It is efficient to look through the tutorial (see Brightspace) before the first contact hours Project assignment: 29/04: 13.15-17:00 (supervised), MIN0.01 01/05: 09.00-12:45 (unsupervised) and 13.15-17.00 (supervised), MIN0.01 M3: Laser and optics Lecture: 9:30-12:30, 29/04, MIN 011; Group assignment: 2 students/group; 12 groups to build 3 different setups and do measurements 29/04: 13-17:00 (unsupervised), MIN 003, 012; 01/05: 09-17:00 (supervised), MIN 003, 012. Supervisor: Hai Wang Module 4. Statistical data analysis Lecture and individual assignment I: 13:15-17:00, 29/04, BBG 079. Lecture and individual assignment II: 09:00-12:0, 01/05, BBG 017. Supervisor: Anuradha Samajdar Module 5. Mass spectrometry Lecture and individual assignment I: 13:15-17:00, 29/04, MIN 0.11. 09:00-12:0, 01/05, BBG 169. Supervisor: Alexis Gilbert Module 6. Proto Lab Lecture and practice time: 9-12:45, 29/04, CBG 0.22; 9-12:45, 01/05, CBG 0.22 Project assignment: In between lectures, at home Supervisor: Pieter Kooijman
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Week 3-9: project phase You will work on a project under supervision and support jointly by a teacher, and a TA. As a group, you will meet your teacher and TA weekly (please discuss it with your teacher about the schedule), and daily supervision will be done via TA.

Supervision and TA support. You will have TA support during the course. TAs are your first contact persons, and they will provide help to you in the lab and can further guide you to the right person (e.g. technician/teachers).

Assignment: In week 4, **by 15/05**, the group is required to submit a brief, max 3 page proposal including: (1) the aim of the projects; what is the question to be addressed? (2) your proposal how to solve the problem; and (3) how you will distribute your task, and how you will work together as a team? You can include teacher's and TA's suggestion and feedback on these aspects

Week 10: Poster submission, poster session and reflection.

Assignment: Students will submit their poster to Brightspace on **22/06/2026, at 12:00 noon**. The coordinator will take care of printing out the posters submitted in time.

Poster session (MIN Learning Plaza, 13:00-16:00 on 24th of June)

You and your group members will stand together in the front of your poster to show what you have done with your group. It is a two-hour festival-like activity with two sessions; half of groups will present, and the other half will have time to visit others' poster.

A poster example or module will be provided to guide the general format.

Reflection and feedback. At the end of the project, ideally in week 10 after the poster session (25-27/06), you will reflect on the project together with the project group and the supervisors of the project, the results obtained on the research question and the cooperation in the project group.

What do we expect from you?

- You work together with the other students in the project group, divide the tasks among yourself and communicates your progress within the project group.
- Together with the project group, you design your experimental project based on the research question and develop a data acquisition system, with which these experiments can be carried out.
- You will use the knowledge you have gained in the modules of this course in the design, analysis, and interpretation of the experimental project.
- You carry out the experiments with the project group, gather the results with the data acquisition system in a scientifically sound manner and archives the results in a transparent way.
- In the project group, you will analyze the results of the experiments and investigate the possibilities to improve and/or broaden the experiments.
- In the project group, you try to rationalize the results of the experiment carried out with a theoretical/numerical model (if necessary).

- You will document all the progress and results of the project in a common, electronic documentation system, which is also open to your supervisors.
- Together with the project group, you will present the results of the research in a public meeting by means of a poster and can individually share all the results achieved.

Evaluation. We will grade you by: **Pass or Fail**. You will receive feedback for your performance based on the **Rubric** which you can find in the latter part of the document.

To pass the course, you need to have all three passes from (1) module, (2) project execution, your lab journal and work attitude (3) poster session, based on the **Rubric** attached.

3.2 Guides to TAs

ERP – TA task description

Considerations

In an ideal situation, a teacher would supervise the projects directly. This would lead to the most effective communication, knowledge transfer and grading. Although we leave teacher's the freedom to choose for direct supervision, in general this model is too time intensive, given that every teacher will supervise up to 3 student groups.

Communication issues are a risk when leaving all coordination up to the student project team with limited direct support from the PI. Experience with the Experiment Design course at the Master level shows that students do not know who to approach in case of which problems and approach no one, or approach multiple parties simultaneously (PI, technical staff, PhD's). This leads to miscommunications and communication effort on all parties involved. We therefore suggest a model where the TA is the intermediary between students and advanced sources of information and the first point of contact.

Besides providing the supervision support during the project phase in week 3-9, TA's will also supervise and help to grade the modules in the second week. We appoint 'Advanced TA's' (ATA), ideally one per institute, preferably a PhD but a MSc student might also qualify. These ATA's will perform general TA tasks (perhaps supervising slightly less students to compensate) but are also available for regular TA's for questions since they have expertise on specific setups or experimental techniques or on doing research in general. Where possible, the ATA is connected to the course for multiple years.

Before the start of the course, there will be **two sessions of TA preparation** (on 7th and 14th of April, 15:30-17:00) in which we aim to provide TA sufficient information on when/how/what to supervise students.

TA tasks:

1. Help preparing module (the module is in week 2 while the preparation work is in week 1)

2. Supervise and help to grade modules in week 2.
3. Provide daily supervision to students (3 to 4 project groups) in the project phase, thereby ensuring:
 - a. Safety in the lab room (presence during contact hours, required for possibly hazardous setups)
 - b. Opportunity for discussion while working on the project
 - c. Resolving low-level practical issues

TA's are expected to be present during regular contact hours, but they are free to make other arrangements with project groups to schedule other contact moments or give on-the-fly support.

4. Have regular meetings with teacher, ATA and/or technical staff, and approach these in the case of questions, unclarities and practical issues. Once the TA distribution to projects/teachers
5. Help to grade (together with teacher) the practical learning goals / components in the rubric.

The table below contains a suggestion for how to put in the 12 hours per week.

TA	Week	Preparation	Supervision	Grading	Total
	-2	Training 6;			6
	1	modules 10 (for week 2)			10
	2		Module 8	Module	8
	3 to 9	Discussion 2	Project 10		84
	10	Poster 2	Poster 4	PI/ATA 6	12
					120

Modules

In the modules, TA work is largely as in the standard bachelor lab courses. Preparing and supervising the TA's during the modules is the responsibility of the instructor / STA for the module.

- TA's can prepare based on lab instructions with 'standard' material (week 1).
- Supervising the modules means being present during contact hours, help with practical issues, and monitoring progress (week 2).
- About helping grading: assuming that each module contains preparatory assignments and a large Final assignment, an efficient option would be:
 - to mark preparatory assignments for completion or leave checking preparatory assignments completely
 - to help in grading the Final assignment based on the rubric (Insufficient / Sufficient / Excellent) with the focus on giving lots of feedback.

Projects

In projects, the TA task is a difficult and vulnerable one. Even more than in standard lab courses, they have a lack of research experience and relevant knowledge (physics, experimental

techniques). In the chosen model, making TA's successful in their role is crucial to the quality of student projects (and thereby to the quality of the course entirely). To make TA's successful, it is important that sufficient organizational support is given:

1. PI's should give TA's relevant information before the start of the project (papers, written / oral instructions, other background material). This is especially effective for predefined projects.
2. TA's should have helplines during the project: they should know who to approach (course coordinator, teacher, ATA, technical staff) in case of a certain type of problem.
3. TA preparation and training should focus on other things than the standard preparation for lab courses and modules. Two TA training sessions will be offered 2 weeks before the course.

TA's will be distributed over projects based on prior knowledge (e.g. TA's supervising the Data acquisition module supervise Arduino projects) and personal interest.

Preparing and training TA's

The TA preparation for projects is partly the responsibility of the course coordinator, and partly of the teacher.

The course coordinator will give a general instruction that focuses on (in 7th and 14th of April; 15:30-17:00):

1. What is expected from TA's before, during and after the project,
2. How to supervise (semi-)open projects,
3. The importance of communication and the lines of communication,
4. The attitude that is expected from them.

This instruction can also be (partly) given to students, so that they know what to expect from their TA. Usually, students see the TA as the source of 'right answers', i.e. a person who can resolve their issues so that they can proceed with their experiment or exercise.

In this course, the TA is more a discussion partner helping them not to resolve issues, but find a way to resolve issues - to find out what needs to be done and where to find the relevant knowledge. When looking at the Perry scheme, what we effectively try to do is introduce the TA into the project as a person who is ahead of them in the scheme and helps them make the transition. It is important that students realize that the TA does not know everything, first of all because they are not hired as experts, and second because also the role we expect from students themselves changes.

To help harness TA's in their complex role, teacher's can

1. Draft a document describing typical things that students will encounter, who to contact when a specific situation occurs, ...
2. Meet regularly (say half an hour per week) without students being present to discuss both the contents and the progress of projects,
3. Be available for questions during the project phase,

4. Step in and contact the course coordinator if practice shows that the TA is not up to the task.

ATA Tasks

1. Be available for TAs for expertise on the physics of the experiment and the operation of setups and instrumentation
2. Provide daily supervision to students (3 to 4 project groups) in the project phase (see TA tasks)
3. Have regular meetings with teacher, TA and/or technical staff, and approach these in the case of questions, unclarities and practical issues
4. Help in grading (together with teacher and TA) the practical learning goals / components in the rubric.

When ATAs are not involved during the modules, they have more time available to help TAs/students in the project phase. We have the following suggestion to distribute the ATA hours:

For ATAs who are involved in module preparation and supervision:

ATA	Week	Preparation	Supervision	Grading	Total
	-2	Training 6;			6
		modules 10			
	1	(for week 2)			10
	2		Module 8	Module	8
	3 to 9	Discussion 3	Project 13		112
	10	Poster 2	Poster 4	PI/ATA 8	14
					150

For ATAs who are NOT involved in module preparation and supervision:

ATA	Week	Preparation	Supervision	Grading	Total
	-2	Training 6;			6
		Modules 0			
	1	(for week 2)			0
	2		Module	Module	0
	3 to 9	Discussion 3	Project ~15.7		~112
	10	Poster 2	Poster 4	PI/ATA 8	14
					150

3. 3 Guide to the teachers

Aim of the course: The new course Experimental Research Project (or shortly, ERP) aims at providing students the opportunity to go through the research cycle once during their bachelor studies. In the course the students should work in a group of 3 and you will be supervising up to 3 groups. The number for steps that they ideally should take are: designing their own research question, designing an experiment that researches that question, constructing the experiment,

taking the data using their equipment, analyzing the data, and last but not least, reporting on the results. Since it will be the first time for them that they will go through the cycle, it will be your task as a supervisor to assist them in all the necessary steps. Research question: The most problematic step for the students will be to design their own research question. In particular, it has to be a research question that can be answered in a period of 10 weeks, and where each student will spend about 200 hours. There are different scenarios possible. If they have an original idea, you can explore it with them and come up with a research plan. If not, you can help them by discussing possible questions that you find interesting. At the end of initial phase of the course, thus after 3 weeks, they should have a viable research plan ready and should be able to start the design of the experiment. Designing the experiment: Depending on the research question, the students should build up their experiment. In this stage you can get them into contact with TA's or technicians that can help them in this phase. It is critically important that they evaluate as soon as possible the components that need to be acquired (bought, or machined), since this should not take too much of their time. Performing the experiment: For this part of the course, the students have ample experience from the first- and second-year courses that have a "practicum" component. If they perform their experiment in the "practicum zalen" in the Minnaert building, there are technicians available that will watch over them. If the experiments are performed in your own lab, assign a technician to guard them. Presentation of the results: In the end, the students will report on their findings in a poster session. This will be a festive activity, but at the same time a learning experience for them. Each student should be able to present the results of the whole project, and in the end each student will be evaluated individually. Task of the supervisor: It is your task to guide the students throughout the course. You will plan together with a TA that has been assigned to them, a short, weekly meeting with the students to discuss their progress. Using a rubric, you must assess their performance in the course on an individual basis. Although the students will only get a pass/fail at the end of the course, the assessment on the various parts of the rubric will help the students to establish an opinion about their performance in the course. In total, the teaching load for you in this course will be around 30 hours. Final meeting: At the end of the course, you should organize a meeting with all the stakeholders involved (students, TA's, technicians) evaluating the project, the results, discussing what went OK, what went wrong, and how they worked together in the team. The results of this meeting should be part of your assessment of the students.

4. Information for general lectures and Modules in week 1-2

To enhance the skills required for conducting a short experimental study within the course ERP, the students have to follow the General Lectures, and 1 assigned Module based on the project selected.

In week 1, we have one information session about the project, and two General Lectures of 4 contact hours.

Lecture 1 (9-12:30 am, 22/04/2026; MIN 001-004)

- Introduction to all projects provided by the coordinators and some teachers from different institutes.
- In-person contact to communicate with teachers directly to find out possibilities and details.
- Nothing for you in the project list? Maybe it is the right place/time to discuss open projects from yourself with teachers

Lecture 2 (13:15-15:15, 22/04/2026; KBG Atlas)

- The scientific method, the research cycle
- Designing an experiment, boundary conditions
- Project implementation

Lecture 3 (9-11 am, 24/04/2026; KBG Atlas) How to work in a team and how to collaborate with others – There is an assignment (with deadline on 15th of May linked to the lecture 3 -so it is important to join it)

Modules (29-31/04/2026, slot A and D; week 2)

Module 1. Electronics

All physical entities are nowadays translated into electric voltages or currents as they are measured. For this reason, it is of utmost importance that all students have a basic comprehension of electronics. That includes the distinction between DC and AC circuits, and the influence of resistances, capacitances, and impedances. Electronics typically perform the first steps of (analog) signal processing, such as low-bandpass and high-bandpass filtering, noise cancelling, coincidence measurements, etc. Analyzing electronic signals in the frequency domain rather than in the time domain will be treated (e.g., think of music as a collection of tones), as the wave character of signals is omnipresent in physics.

Lecture time: None, preparation at home (see instruction manual on Brightspace)

Project assignment:

29/04: 13.00-17:00 (supervised), MIN002; 01/05: 09.00-13:00 (supervised) and 13.00-17.00 (unsupervised), MIN 002

Supervisor: Marijn van Huis

Module 2. Data acquisition and sensors (ARDUINO)

The topic of this module is the data acquisition of the output of a sensor measuring e.g. temperature, pressure, humidity, movement, or concentration of particular gases (methane, carbon dioxide, NO₂, etc), building the setup required for the data acquisition, and programming the hardware-software interface in such a way that it allows for computerized processing and analysis. To this end, commercially available devices are available that are called ARDUINO chips (see, e.g., <https://robocraze.com/blogs/post/arduino-sensor-types-and-applications>). Some programming in Python will be required to extract data from the self-built electronics setup including the ARDUINO boards. A tutorial will be provided to guide you step-by-step through this process.

Lecture time: None. It is efficient to look through the tutorial (see Brightspace) before the first contact hours

Project assignment:

29/04: 13.15-17:00 (supervised), MIN0.01

01/05: 09.00-12:45 (unsupervised) and 13.15-17.00 (supervised), MIN0.01

Supervisor: Peter van Capel and Giogio Cover

Module 3. Laser and Optics

In this Module, several aspects of optics, and lasers and laser safety, optical elements, and general principles of how to align a setup will be discussed in the lecture. Then we will have 2 people/group to develop one of 3 setups that you will assign to: interferometer; polarization experiments and diffraction.

Lecture time: 9-12:00, 29/04, MIN 011;

Group assignment: 2 students/group; 12 groups for building 3 different setups for measurements

13-17:00, 29/04 (unsupervised), 01/05; 09-17:00 (supervised), MIN 003, 012.

Supervisor: Hai Wang

Module 4. Statistical data analysis

In this Module, statistical analysis of data and output signals will be treated, depending on the underlying physical model or phenomenon. Statistical (peak) analysis is of pivotal importance in many physics experiments such as in the discovery of the Higgs boson at CERN, and various

modelling approaches and algorithms can be used to determine whether measured signals are significant or not, and whether these conform with a proposed physical model.

Lecture and individual assignment I: 13:15-17:00, 29/04, BBG 079.

Lecture and individual assignment II: 09:00-12:00, 01/05, BBG 017.

Supervisor: **Anuradha Samajdar**

Module 5. Mass spectrometry

In this Module, spectrometry (spectroscopy of particles, ions, fragments) that allow compositional analysis of gases or solid materials, will be treated. The detection technique typically makes use of either the different masses or the different charges of the ions or particles, or both. Also isotope analysis is one application. Sometimes spectrometry can be combined with microscopic methods.

Lecture and individual assignment I: 13:15-17:00, 29/04, MIN 0.11.

09:00-12:00, 01/05, BBG 169.

Supervisor: **Alexis Gilbert**

Module 6. Proto Lab– Digital fabrication

This is a hands-on Module where experimental skills and techniques are taught in the Proto Lab. This includes This 3D design, 3D printing, soldering and laser cutting. Ideally students should choose this Module if they think it will be of use in their projects. Because of a limited capacity in the Proto Lab, there is a maximum number of 25 students who can follow this module.

Lecture and practice time: 9-12:45, 29/04, CBG 0.22; 9-12:45, 01/05, CBG 0.22

Project assignment: In between lectures, at home

Supervisor: **Pieter Kooijman**

5. Project description (for week 3-9; 47 projects in 2026 in total)

5.1 A quick overview on the teacher and project (to be updated for 2026)

Note that: project 19 and 25 are open projects. If you would like to develop a new topic of yourselves, that is possible and please talk to Hai Wang/Marijn van Huis. Please put these two projects as your top 2 wishes if you would like to do an open project.

#	Teachers	Project name
1	Marc Schneider, Erik van Sebille	<u>Measuring water surface movements with a self-built floating GNSS tracking system</u>
2	Marc Schneider, Erik van Sebille	<u>Data collection and analysis for an outdoor GNSS tracking system</u>
3	Carleen Tijm-Reijmer	Meteorological measurements: Temperature, humidity and wind
4	Carleen Tijm-Reijmer	Meteorological measurements: Solar UV/VIS/IR radiation and wind measurements
5	Carleen Tijm-Reijmer	Meteorological measurements: Solar UV/VIS/IR radiation and solar energy production
6	Alexis Gilbert, Thomas Röckmann	Isotopic composition of methane (CH ₄) emitted from different sources
7	Alexis Gilbert, Thomas Röckmann	Natural isotopic composition of carbon dioxide (CO ₂) emitted from different sources
8	Alexis Gilbert, Thomas Röckmann	Isotopic fractionation of carbon monoxide (CO) in photolytic simulation experiments
9	Rupert Holzinger & Nemat Omidikia	Environmental Research Using a Very Sensitive and Specific Mass Spectrometer I
10	Rupert Holzinger & Nemat Omidikia	Environmental Research Using a Very Sensitive and Specific Mass Spectrometer II
11	Allard Mosk	The electro-optics cookbook
12	Allard Mosk	I spy with my little lensless camera
13	Allard Mosk	Roll over Fraunhofer
14	Dries van Oosten	Acoustic experiments
15	Dries van Oosten	Seeing shock waves
16	Dries van Oosten	The physics of musical instruments
17	Peter van der Straten	Speed of Light
18	Peter van der Straten	Saturation spectroscopy
19	Hai Wang/Marijn van Huis/ Peter van der Straten	Open projects or Experimental quantification of plank constant.
20	Hai Wang	Who am I?
21	Marta Verweij	Muon Tomography
22	Thomas Peitzmann	Muon lifetime
23	Alessandro Grelli	Muon Tracking with GEMs
24	Anuradha Samajdar, Chris van den Broeck,	Thermal Expansion with an Interferometer
25	Marta Verweij, Olaf Massen,	Open project
26	Marta Verweij	Build your own pocket-sized particle detector
27	Joost de Graaf	Dynamics in bubble rafts or Spheres adsorbed to a water-air interface
28	Joost de Graaf	Measuring the jamming of colloids in a floating droplet

29	Nadine van der Heijden	Home built Scanning Probe Microscope
30	Nadine van der Heijden	Looking at atoms with the Nanosurf NaioSTM
31	Peter van Capel	The upward driven disk
32	Peter van Capel	Build your own EEG
33	Freddy Rabouw	Neuromorphic computing with photon-avalanching nanocrystals
34	Marijn van Huis	Elektronentomografie van nanodeeltjes
35	Arnout Imhof	Soft crystallography with light
36	Arnout Imhof	When light gets drunk
37	Arnout Imhof and Alfons van Blaaderen	Efficiency of random closest particle packings
38	Alfons van Blaaderen	Measuring pair potentials between colloidal particles
39	Alfons van Blaaderen	Phase behavior of a 2D colloidal model system
40	Gerhard Blab	OCT Data Analysis Project (I)
41	Gerhard Blab	OCT Data Analysis Project (II)
42	Marjolein Dijkstra	Bouwen van neurale netwerken met elektrische neuronen -I
43	Marjolein Dijkstra	Bouwen van neurale netwerken met elektrische neuronen -II

5.2 Project description

IMAU (10 projects)

A. Tracking systems (Erik van Seville, Marc Schneider)

Project 1. Measuring water surface movements with a self-built floating GNSS tracking system

This project includes the setup and assembly of an Arduino-based GNSS tracking + SD card logging + battery system, the conception and construction of a waterproof housing, followed by some outdoor water measurements in the vicinity of the Science Campus (for example on the Kromme Rijn). The used materials are an Arduino MKR board with compatible GNSS and SD card hardware, a battery for power, as well as some material to construct the housing. Lili's Proto Lab can be used to design waterproof housing. For this reason, all group members should complete the Lili's Proto Lab module of the course as preparation.

Project 2. Data collection and analysis for an outdoor GNSS tracking system

This project focuses on working with position data from a GNSS system that is collected around the Science Campus during the project. The data analysis aims at studying the reliance of the positioning system: Understanding its accuracy, environmental impacts such as the presence or absence of buildings. The measurements are collected with an Arduino-based GNSS tracking + SD card logging + battery system in a weatherproof containment, which can be set up and configured with the Arduino IDE. The used materials are an Arduino MKR board with compatible GNSS and SD card hardware, a battery for power, as well as weatherproof housing.

Suggested modules: Arduino; Proto lab; Electronics and signal processing

B. Meteorological measurements (Carleen Tijm-Reijmer)

The idea of these are 3 projects is to develop a simple “weather station” that continuously records environmental variables (temperature, wind, humidity, etc..) using Open Source software and Arduino, such as <https://openweatherstation.com/ows/index.php#weather-data> or the recipe from <https://doi.org/10.1016/j.ohx.2023.e00402> (see also <https://doi.org/10.1890/0012-9623-95.2.68>)

The described 3 projects use a similar general set-up, but differ in their focus on different processes or topics. In all cases a goal is formulated, but other ideas are welcome, though limited by the provided instrumentation.

Project 3: Temperature, humidity and wind

Students will assemble their own water-resistant weather station, with temperature, humidity and wind speed as the key measurement parameters. First different data sampling strategies will be tested. Data will be compared with benchmark sensors and loggers from IMAU which are used in other courses (MAIO). The final goal could be to estimate evapo-transpiration from an area around campus (e.g. Botanical Gardens) using the Penman-Monteith equation.

Project 4: Solar UV/VIS/IR radiation and wind measurements

Students will assemble their own water-resistant weather station, with solar UV/VIS/IR radiation and wind as the key measurement parameters. First different data sampling strategies will be tested. The final goal could be to estimate PV/wind production at different locations.

Project 5: Solar UV/VIS/IR radiation and solar energy production

Students will assemble their own water-resistant weather station, with solar UV/VIS/IR radiation and a solar panel as the key measurement parameters. First different data sampling strategies will be tested. The final goal could be to estimate the efficiency of the solar panel under different conditions.

Suggested modules: Arduino; Proto lab; Electronics and signal processing

C. Isotope measurements (Alexis Gilbert, Thomas Röckmann)

Project 6. Isotopic composition of methane (CH₄) emitted from different sources

In recent years, researchers, engineers, and policy makers have increasingly focused on understanding and reducing methane (CH₄) emissions. Measurements of the stable isotopes of methane (¹³C, ²H) help constrain its sources and sinks in diverse environments. In this project, students will collect air or gas samples at a selected location (farm, ditch, highway, etc.), which will be chosen together with the supervisors. The samples will then be analyzed in the laboratory using isotope ratio mass spectrometry (IRMS). Based on the results, students will interpret the sources (and possible sinks) of methane at this location and discuss the potential limitations and uncertainties of their analysis.

Project 7. Natural isotopic composition of carbon dioxide (CO₂) emitted from different sources

Carbon dioxide is a major component of the atmosphere, and understanding its origin is essential for both local and global climate models. The goal of this project is to measure the stable isotope composition of CO₂ from different environments (forest, lake, highway, etc.) to assess the contribution of various sources to atmospheric CO₂. The sampling location and timeframe will be defined by the students in consultation with the supervisors. Samples will be collected in the field and analyzed in the laboratory using Tunable Infrared Laser Direct Absorption Spectroscopy (TILDAS).

Project 8. Isotopic fractionation of carbon monoxide (CO) in photolytic simulation experiments

In this project, students will conduct photochemical experiments using a photoreactor to simulate the production and degradation of carbon monoxide in the atmosphere. Experiments will be performed under different conditions (temperature, pressure, wavelength, etc.). For each experiment, the ¹³C and ¹⁸O isotope composition of the produced or remaining CO will be analyzed using an isotope ratio mass spectrometer (IRMS). Based on the data, students will (i) develop a chemical model to better understand the isotope fractionation associated with photolytic production and consumption of carbon monoxide and (ii) discuss the implications for natural atmospheric observations.

Suggested modules: Mass spectroscopy; Statistical Data Analysis; Laser and optics

D. Environmental Research Using a Very Sensitive and Specific Mass Spectrometer (Nemat Omidikia and Rupert Holzinger)

Capacity: two projects

Interested in finding out whether sea foam contains nanoplastics? This and many other things you can investigate by choosing for our project! You will use the TD-PTR-MS setup in the IMAU laboratory, which you can think of as a super-sensitive artificial nose.

‘TD’ stands for ‘thermal desorption’, meaning that the sample is heated under defined conditions to a final temperature of 350°C. The vapours that come off are measured by the mass spectrometer. ‘PTR’ stands for ‘proton transfer reaction’ and refers to the ionization method of the vapours. Hydronium, H₃O⁺, is the proton donor and whenever a H₃O⁺ ion collides with an organic molecule a H⁺ attaches to the organic vapour which is thus ionized. Proton transfer is a soft ionization method that preserves the

chemical profile of the sample. The time of flight mass spectrometer (MS) measures the intensity of the ions as well as the molecular weight.

The analysis of a sample takes ~25 minutes typical contains the time rows (1 second resolution) of hundreds of different ions. You will use principal component analysis (PCA) and other machine learning tools to extract information. Nanoplastic measurements can be calibrated by the standard addition method, which involves spiking several replicas of one sample with known concentrations of nanoplastics.

Two types of projects are possible: (i) **Project 9**: analysing environmental samples, or (ii) **Project 10**: further developing the method.

When analysing environmental samples, you start by defining a research question, e.g. does sea foam contain nanoplastics? Subsequently you develop a sampling strategy as well as an analysis and data processing plan.

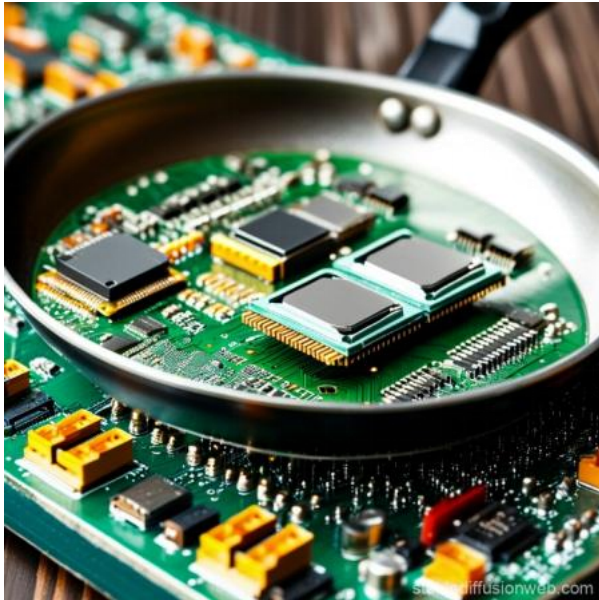
Further method development is always possible. For example, liquid samples need to be filtrated before the analysis because larger particles need to be removed. But to what extent are nanoparticles as well removed in the filtering step? A small study can provide first answers to this question. Method development also includes new applications. For example, our artificial nose could be used to determine whether food is spoiled – a first method of such an application can be developed. Many other applications are thinkable and can be explored as project in this course.

Suggested modules: Mass spectroscopy; Statistical Data Analysis;

Nanophotonics (10 projects)

Allard Mosk

Project 11: The electro-optics cookbook



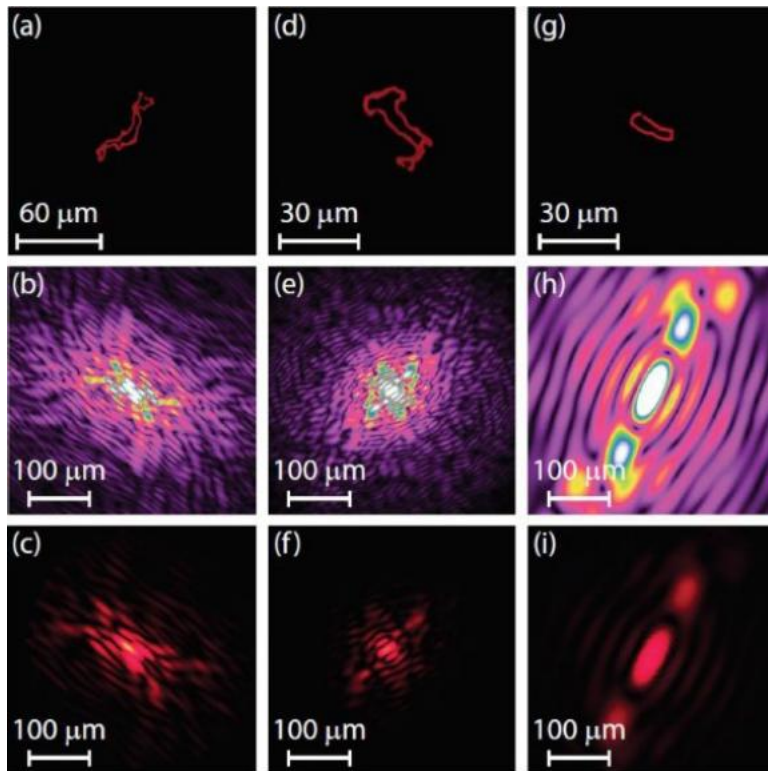
The behaviour of electronic components changes with temperature. In this project, we are going to find out.

- Study the theory of semiconductor based opto-electronics
- How does the sensitivity of a camera to different wavelengths of light change with temperature?
- What about the emission wavelength of a LED or a (cheap) laser diode? How does it relate to the change in the IV curve?
- Can we use this to estimate Planck's constant?
-

Contact: Allard Mosk, Nanophotonics, a.p.mosk@uu.nl

-

Project 12: I spy with my little lensless camera



Three countries, their calculated Fourier transforms, and measured diffraction patterns in the Fraunhofer regime. (Source: A. F. Rigosi, Analysis of Fraunhofer diffraction patterns' entropy based on apertures shaped as national borders, <https://doi.org/10.1016/j.ijleo.2018.07.133> .)

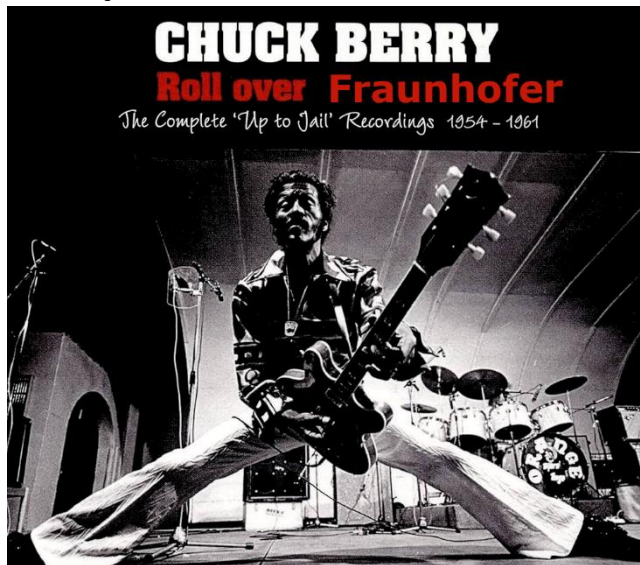
In the Fraunhofer diffraction regime, the diffraction pattern of an object is essentially its Fourier transform. Let's test this! Using some slides that contain letters and other easily recognizable objects, we study their Fraunhofer diffraction patterns and compare them to simulations.

In many cases, it is possible to recover the object from the Fraunhofer pattern. That is, we can have a camera without a lens! If we illuminate an object with a laser we can even try to see its Fraunhofer diffraction pattern even through non-transparent "privacy glass".

Here we can test:

- How closely do the measured diffraction patterns resemble the theoretical ones?
- How to measure similarity of images anyway?
- What algorithms exist to retrieve the object from the diffraction pattern?
- What is the success probability in practice

Project 13. Roll over Fraunhofer



In this project we will study signal processing and data compression of audio signals.

First, we need to record a bit of music (or put on an old Chuck Berry record). Then, we explore several signal processing choices and data compression strategies.

- Why do we need filters?
- How to optimally reconstruct a sampled signal?
- The popular Fraunhofer-MP3 algorithm is based on a psychophysical effect called masking. How do you measure masking? Can you experience it and gather evidence from a range of volunteers (your fellow students and professors)? Does it depend on age? Do people who practice music (singing, playing an instrument) experience different levels of masking than people who do not? Does it correlate with hearing acuity?
- Psychosensory effects such as masking are notoriously hard to measure. Can you find a way to quantify the reliability (or lack thereof) of your result?

Dries van Oosten

Project 14. Acoustic experiments

In some of the introductory courses you have taken, such as Relativistische and Klassieke Mechanica and Golven en Optica, you have learned about the similarities and the differences between sound and light. An important practical difference between is that with light, we cannot directly measure the oscillations, as the frequency is much too high (hundreds of THz's) and the wavelength is extremely small (hundreds of nm's) whereas acoustic vibrations can be easily measured by a simple microphone. Using a modern, widely available usb-audio interface, we can in fact sample many channels

simultaneously. Doing experiments with acoustics is therefore very illustrative and it allows us to explore many techniques that are difficult to implement with light. In this project, we will

- study sound propagation, diffraction and interference.
- use sound to do ranging and detection of objects (sonar)

Contact: Dries van Oosten, Nanophotonics, D.vanOosten@uu.nl

Project 15. Seeing shockwaves in air

When an object moves through a gas, the pressure in front of the object increases, causing the gas to be pushed aside. Changes in pressure propagate through the gas with the speed of sound, so what happens when the object moves faster than the speed of sound? The pressure won't increase in time for the gas to be pushed aside. Nature's way of solving this problem is by creating a shockwave. In this project, we will study the behavior of such shockwaves. However, to do this, we have to be able to see them! For that, we will make use of a clever trick, called Schlieren imaging. In this project, we will

- develop a Schlieren setup and figure out how to optimize the contrast
- study how supersonic gas flows expand, based on the shape of the nozzle
- study how supersonic gas flows interact with obstructions in the flow and how this depends on the shape of the obstructions.

Contact: Dries van Oosten, Nanophotonics, D.vanOosten@uu.nl

Project 16: The physics of musical instruments

The sound produced by a musical instrument contains many subtleties that determine its timbre. We can all hear the difference between an electric and an acoustic guitar. We can even hear the difference between guitars with steel or nylon strings. At first glance, the timbre of an instrument is determined by the (harmonic) spectrum of the instrument and its envelope (i.e. how fast the

amplitude decays as a function of time). However, the story is a bit more subtle. When we talk about the harmonic of a note, we are already assuming that the overtones of the instrument are in fact harmonic (i.e. they have a frequency that is an integer multiple of the fundamental frequency of the note), which they are often not. Also, the decay time of each overtone is in fact different. High overtones generally decay more quickly than low overtones. In this project, we will

- investigate the overtone spectrum of an instrument of your choice
- investigate how the different overtones decay as a function of time

· attempt to identify the dominant decay mechanisms. Is this just things like friction, or is a significant amount of energy carried away as sound?

Contact: Dries van Oosten, Nanophotonics, D.vanOosten@uu.nl

Peter van der Straten

Project 17. Speed of Light: The speed of light is a fundamental constant in Nature. Measuring the speed of light with a large precision is a delicate experiment. Here the students will measure the speed of light by using the beating of longitudinal modes in a HeNe laser. By changing the length of the open-cavity, the shift of the cavity modes can be exploited to find a rather good measurement of the speed of light. The students will setup the experiment and detect the speed of light in air.

Project 18. Saturation spectroscopy: Spectroscopy of atoms and molecules is hampered by the Doppler effect, where the motion of the atoms or molecules produces shifts of their resonance frequencies that is well beyond the natural linewidth of the transition. Using a clever scheme for the interrogating lasers, transition lines can be observed that are well below the Doppler width and are of the order of the natural linewidth. The students will setup sub-Doppler spectroscopy and observe the hyperfine shifts in the lowest resonant transition of rubidium atoms.

Project 19. Open project (Marijn van Huis/Hai Wang)

or

Experimental quantification of plank constant (Hai Wang/ Peter van der Straten).

Let us build a time machine and travel back to 1905: Einstein sent a paper, titled "On a Heuristic Viewpoint Concerning the Production and Transformation of Light," to the German journal *Annalen der Physik* on March 17, 1905. Einstein proposed that light consists of discrete energy quanta (photons) whose energy is given by $E=h\nu$, where h is Planck's constant and ν is the frequency of the light. When light strikes a metal surface, an electron is ejected only if a single photon carries enough energy to overcome the metal's binding energy (work function, ϕ), expressed by the equation $h\nu=\phi+(1/2)mv^2$. This explained why the kinetic energy of emitted electrons depends linearly on frequency, not intensity, and why a threshold frequency $\nu_0=\phi/h$ exists below which no electrons are emitted—a phenomenon classical wave theory could not account for.

Can you help Einstein to experimentally prove his proposal using a setup built by yourselves, and quantify the plank constant as the side product?

Elective Modules: optics, electronics, Protolab

Hai Wang

Project 20. Who am I



I have two crystalline materials, who have forgotten who they are. Can you solve the mystery and help them find their name and ID back?

You're welcome to propose your own solution, but keep in mind that you will need to construct a setup to implement it. One possible approach among several options is to use Raman spectroscopy. When laser light interacts with matter—whether it's a solid, liquid, or gas—most photons bounce off without any change in energy. This is called Rayleigh scattering. But a tiny fraction of the light (~ 1 in

a million photons) does something more interesting: it exchanges energy with the molecules/solids, emerging with a slightly different energy. This shift, known as Raman scattering, reveals unique vibrational "fingerprints" of the molecules, helping us identify their structure and composition.

In this project, you'll design and build a working Raman spectrometer from the ground up. Your mission? Use it to analyze two unknown samples and figure out what they are. Can you solve the crystal mystery? Take on the challenge and find out!

Elective Modules: optics, electronics, Protolab

GRASP (6 projects)

PIs: Marta Verweij, Thomas Peitzmann; Alessandro Grelli; Chris van den Broeck, Anuradha Samajdar, Marc van der Sluijs (20%), Raimond Snellings

Elective Modules

- Muon lifetime: data analysis, data acquisition, electronics
- Cosmic Muon Tracking with GEMs: data analysis, data acquisition, electronics
- Muon tomography: data analysis, data acquisition, electronics
- Thermal Expansion with an Interferometer: data analysis, optics, data acquisition
- Build your own pocket-sized particle detector: data acquisition, data analysis, proto lab
- Open project: data acquisition, proto lab, electronics

Project 21. Muon Tomography

In this project you will use cosmic muons to make images of dense materials. Cosmic muons are the result of high-energy radiation that originates from outside our solar system. When they reach Earth, the rays collide with particles in the upper atmosphere to produce a “shower” of particles, including muons. These muons hit the surface of the earth and can be measured. The aim of this project is to use these muons to study materials by measuring them before and after they traverse the material. This is achieved by using a table-top experiment using scintillators in which the muons will leave a signal that can be collected using a photomultiplier tube. With a mechanical robot the setup can also do a scan. For this project students will work on the experimental setup, select materials to be scanned, write code to steer the setup, analyze the collected data, and explore novel applications of this technique. The images that are obtained can be fed to machine learning algorithms to aid in finding the patterns of interest. To get a better idea of this project, you can have a look at this repository: <https://github.com/mverwe/Top-Muon-Tomography>

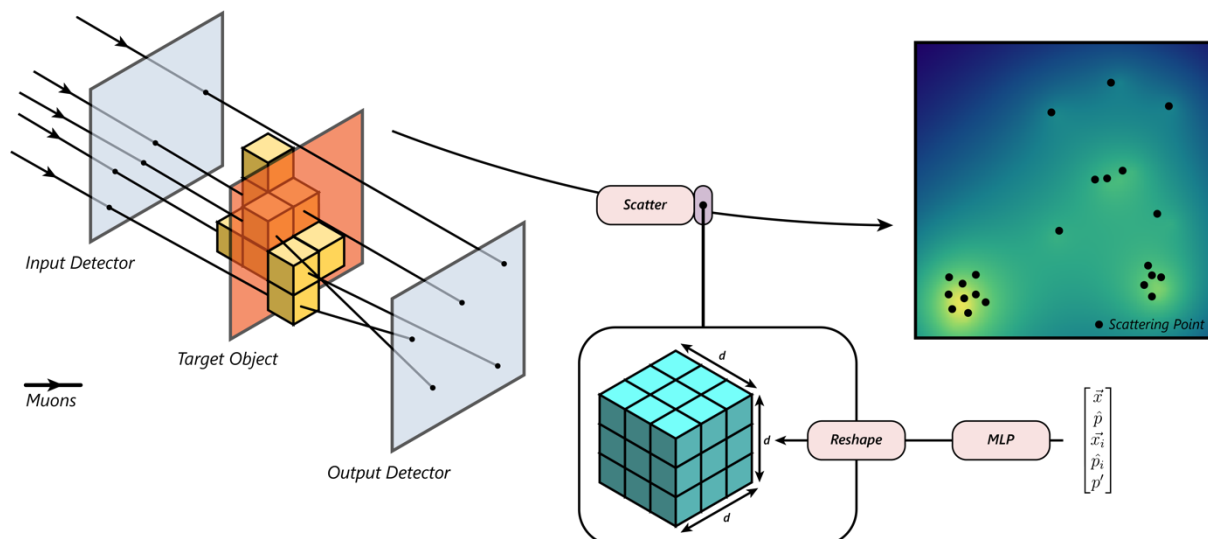
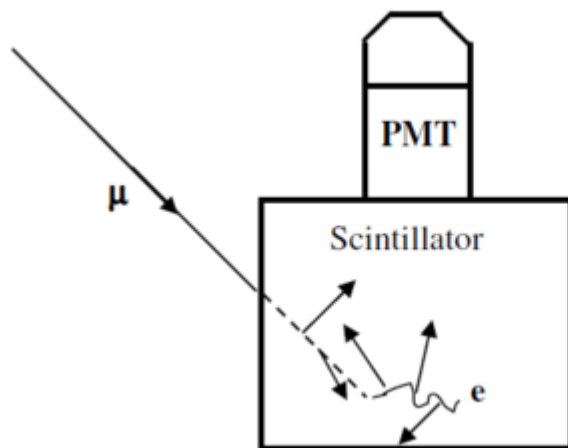


Image source: <https://github.com/jedlimlx/Muon-Tomography-AI>

Contact: Marta Verweij, GRASP, m.verweij@uu.nl

Project 22. Muon Lifetime

The goal of our experiment is the measurement of the lifetime of muons produced in cosmic ray air showers. We will review the theoretical background and discuss the principle of the measurement. The general idea is to obtain a signal for the arrival of a cosmic muon, and after that a signal from the products of the muon decay, and measure the time difference between the two. All physics signals will be generated in plastic scintillators connected to photomultiplier tubes. We will learn about the basic instruments we will use in the measurement (scintillators and PMTs, processing electronics and a digital oscilloscope) and make ourselves familiar with them. We will then assemble the main setup, perform the measurement and analyse them to extract the lifetime.



Contact: Thomas Peitzmann, GRASP, T.Peitzmann@uu.nl

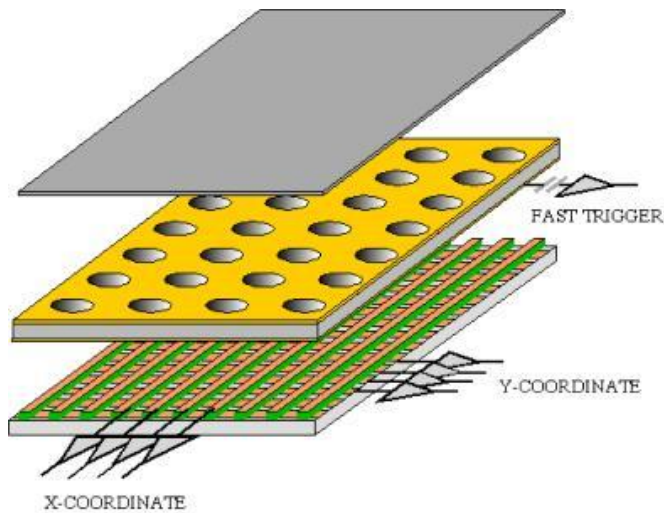
Project 23. Cosmic Muon Tracking with GEMs

The goal of this project is to define a trigger strategy for a Gas Electron Multiplier (GEM) detector when a cosmic muon passes through it.

A GEM detector is a gas detector with an electron multiplication plane to focus and speed up the signal. Those kinds of detectors are employed in the ALICE experiment at CERN.

To study the GEM detector performances, it is imperative to know when a subatomic particle (a muon) pass through it so to be able to distinguish real signal from uncorrelated electronic noise. Such knowledge is given by the so-called trigger system. Usually, a trigger is built with faster detectors placed in optimal positions and proper logical circuitry to give an early warning that a real muon is passing through the GEM

The project consists in building an optimal trigger system for the GEM detector and in investigating its performance versus different possible configurations. To this end the students will work with scintillator detector and nuclear instrument modules, will build the setup, acquire and analyze data.

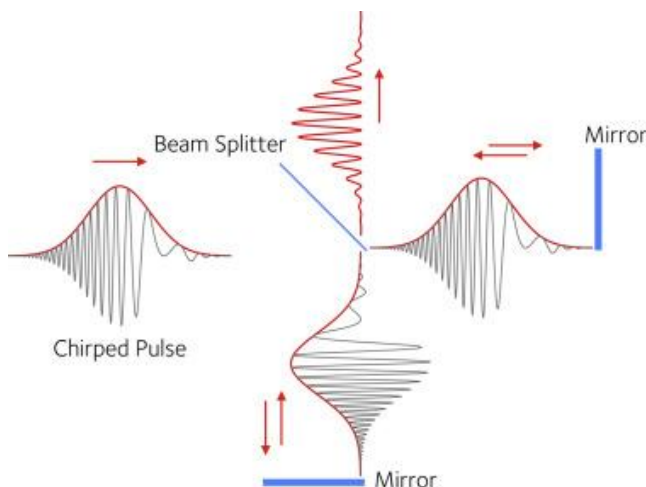


Contact: Alessandro Grelli, GRASP, a.grelli@uu.nl

Project 24. Thermal Expansion with an Interferometer

An L-shaped interferometer (IFO) operates by splitting a laser beam into two paths, sending the beams along perpendicular arms of equal length. At the end of each arm, mirrors reflect the beams back to the beam splitter, where they recombine and produce an interference pattern. Tiny changes in arm length result in detectable shifts in the interference fringes—this principle underlies the high-precision measurements used in gravitational wave detectors like LIGO, which achieved the first direct detection in 2015.

In this project, students will set up a tabletop IFO and use it to detect minute length changes by observing fringe shifts with a photodiode. One of the IFO arms will be made of aluminium and slowly heated during the experiment. As the temperature rises, thermal expansion will cause the aluminium arm to lengthen, shifting the mirror and producing a measurable change in path length. By monitoring these shifts and analyzing the data using Bayesian methods, students will estimate the coefficient of thermal expansion of aluminium.



Contact:

Anuradha Samajdar, GRASP, a.samajdar@uu.nl

Chris van den Broeck, GRASP, c.f.f.vandenbroeck@uu.nl

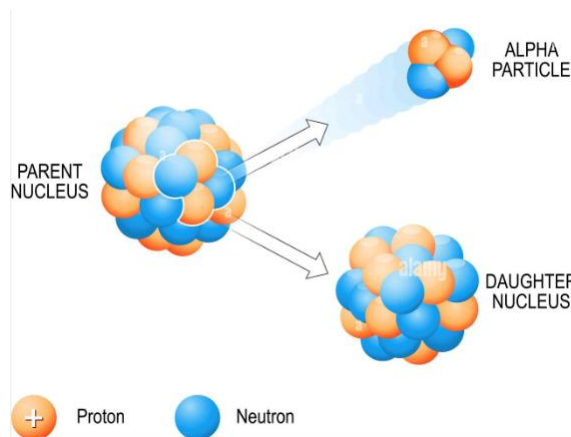
Project 25. Open Project

Come with your own idea using a sensor and an Arduino. Build your own experiment from scratch to find answers to your questions. We will brainstorm with you and help you in each step of the project.

Contact: Marta Verweij, GRASP, m.verweij@uu.nl, Olaf Massen, GRASP, o.massen@uu.nl

Project 26. Build your own pocket-sized particle detector

Experimental particle physics is known for building multiple meters large particle detector at accelerators far underground such as CERN. The weak force however can also be studied with a particle detector that fits into the pocket of your jeans. In this project students will build their own pocket-sized detector to measure subatomic particles. The device they will engineer will be able to measure the energy spectrum of alpha particles. The student group will define their own research plan and decide on which radioactive elements to measure. Besides building the device, the project also entails calibrating the device, research the optimal conditions to perform measurements, and design the data acquisition setup. The group of students is expected to determine up to which extent their device can differentiate between various radioactive materials.



Contact: Marta Verweij, GRASP, m.verweij@uu.nl

Jullius Insistute (6 projects)

Joost de Graaf

Project 27. Dynamics in bubble rafts or Spheres adsorbed to a water-air interface

Dynamics in bubble rafts: Students need to learn to create stable bubble rafts. Will be supplied with aquarium pump and containers. They have to examine a paper about the dynamics in one of these rafts and try to reproduce this. This means learning how to make good bubbles, building a few small

components, analyzing images taken using a webcam, identifying dislocations, etc. The project can go in the direction of characterizing reciprocal dynamics under shear, how bubbles flow around certain geometries, how rafts behave on non-flat geometries, etc. A reasonable source to start with should be: https://www.doitpoms.ac.uk/tlplib/dislocations/dislocation_motion.php

Spheres adsorbed to a water-air interface: Students need to learn about interfacial absorption and capillary effects, as well as how the resulting structures respond to rheological perturbation such as shear. They will learn how to analyze these structures, especially when they are disordered, and what basic rheological measurements involve. They will be supplied with some containers and millimetric plastic beads, but will need to make some small components to do the shear experiments. One thing they could look at is the onset of irreversibility and what the effect of the Reynolds number is. I am thinking something along the lines of:

<https://pubs.rsc.org/en/content/articlehtml/2020/sm/c9sm02482d>

Project 28. Measuring the jamming of colloids in a floating droplet

In this experiment an acoustic trap is used to hold a droplet of particle suspension levitated in the middle of air. The sample could, for example, be suspended colloids (starch) or oil droplets in water. The droplet deformation oscillations are continuously measured with a laser probe and can be analysed to extract the droplet viscosity. The goal of this project is to determine the viscosity of the droplet as a function of volume fraction by evaporating the suspending medium during the experiment. The viscosity is expected to vary drastically at the jamming transition. This could be used as a proof-of-principle for a one-shot measurement that determines the volume fraction at which the system jams. Intrepid students can vary on the setup to help accelerate drying and/or consider how kneading the droplet using higher frequency soundwaves impacts jamming.

Nadine van der Heijden,

Julius Institute,

n.j.vanderheijden@uu.nl

Project 29: Home built Scanning Probe Microscope

Building a Scanning Tunneling Microscope (STM) or Atomic Force Microscope (AFM), or a large-scale analog is a challenging project, but one that does result (hopefully) in a tangible object.

Some examples:

<https://doi.org/10.1119/5.0094028>

<https://doi.org/10.1119/1.16703>

<https://doi.org/10.1119/10.0001700>

<https://doi.org/10.1119/1.4941929>

<https://dberard.com/home-built-stm/>

Proposed modules: Electronics, Arduino, Proto Lab

Project 30: Looking at atoms with the Nanosurf NaioSTM

In this project you learn to use the NaioSTM. Aside from the standard imaging mode and $I(Z)$ curves, you will look at more interesting or challenging surfaces and develop measurement- and data analysis techniques to characterize a surface.

<https://www.nanosurf.com/en/products/naiostm>

<https://journals.aps.org/prl/pdf/10.1103/PhysRevLett.50.120>

<https://journals-aps-org.utrechtuniversity.idm.oclc.org/prb/pdf/10.1103/PhysRevB.58.R4266>

Proposed modules: Electronics, Arduino, Statistical data analysis

Peter van Capel,

Julius Institute,

p.j.s.vancapel@uu.nl

Project 31: The upward driven disk

An upward driven disk (UDD) is a mechanical system consisting of a flat disk that is constantly driven upward by a rotating rubber wheel. Once the center of gravity of the disk gets above the rubber wheel / point of suspension, the disk can fall. The dynamics of the disk can be described by a set of differential equations. An analysis of video measurements show that the disk can display various types of motion, including chaotic behaviour. A recording and a detailed description can be found via the following links:

<https://www.youtube.com/watch?v=PkyFFNbrUI4>

<https://doi.org/10.1119/5.0147893>

There are two setups available: the LEGO-version described in the links, and a slight variation of this setup. This project allows for several variations. The starting point is to build and use Python (real-time or delayed) tracking software that allows for the tracking of the motion of the disk, and calibrate one setup or both. Then, several research goals are possible (these are suggestions, students are free to pursue other goals).

1. It is interesting to perform similar experiments with both setups to test whether the behaviour is comparable.
2. Actually preliminary measurements have shown that the characteristics of the friction between the disk and the driving wheel is different for both setups, and that this has an effect on the movement patterns of the disk.
3. A new digital camera with higher frame rate has recently been bought. Students can implement this camera and investigate whether the quality of the video analysis improves.

4. As suggested in the linked paper, it is possible to perform measurements with a different shape of the disk. Every desired disk shape can be made with the laser cutter in the Proto Lab.

Proposed modules: Proto lab, Data processing, Electronics

Project 32: Build your own EEG

In ERP 2025, one of my groups started building a EEG (Electro-EncefaloGram) device to measure brain waves. Although all materials are selected and bought, getting the device to work proved to be a challenge.

The 'setup' consists of basically three parts: one or more sensors with that can be attached to the head (integrated in a cap), analog electronics to amplify signals and filter unwanted frequencies, and data collection (either done with Digilent Analog Discovery 2 (AD2) Waveforms or automated via Arduino). The main challenge was to build an electronic band pass filter to select just the relevant frequency range corresponding to the type of brain wave.

We will supply you with all materials (sensors, electronics and notes) of the group of last year, which can be used as a starting point.

<https://en.wikipedia.org/wiki/Electroencephalography>

<https://nin.nl/about-the-brain/brain-waves/#>

Proposed modules: Electronics, Data processing, Arduino

SCMB (12 projects)

Freddy Rabouw

Onderzoeksgroep: Soft Condensed Matter & Biophysics,

Contact: Dr. Freddy Rabouw, Email: f.t.rabouw@uu.nl

Project 33: Neuromorphic computing with photon-avalanching nanocrystals

In this project we are going to use “photon-avalanching nanocrystals” for energy-efficient computer operations. These nanocrystals emit light under illumination with laser pulses, similar to conventional fluorescence. What makes photon-avalanching nanocrystals special, is their memory functionality. A laser pulse at $t = 0$ produces weak output light, but a second laser pulse shortly after (within <50 ms) produces stronger output. The nanocrystals store some energy of the first laser pulse in a dark state, which makes the response to subsequent laser pulses stronger. They thus “remember” the first laser pulse. This allows, in theory, for energy-efficient computer operations.

The ERP students will operate optical equipment and perform data analysis. The figure below is from a research proposal showing a potential proof of concept. The ERP students may study the memory functionality on a fundamental level and/or attempt projecting spatial patterns as in the figure.

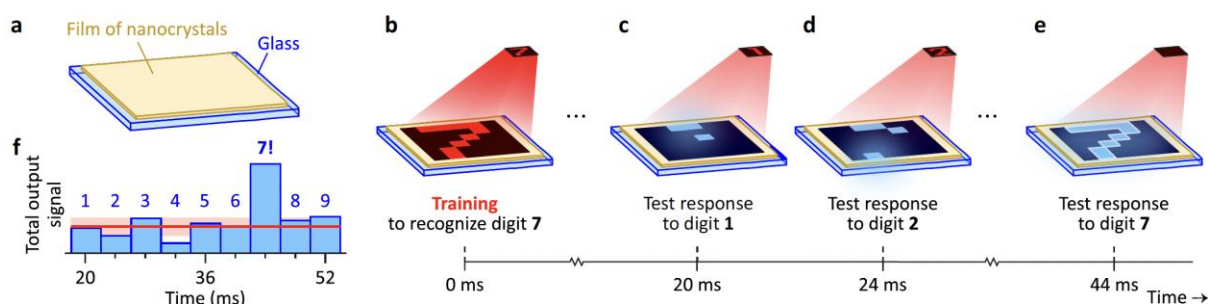


Figure 2 | Energy-efficient image recognition. (a) A film of nanocrystals will act as memory and a signal processor simultaneously. (b) An image of a digit will be projected onto the film—in the example, a digit 7. This will **train nanocrystals** in the illuminated regions to later produce increased output. (c,d) Next, we will use the film for **image recognition**. Test images of all digits 1–9 will be (weakly) projected onto the film successively. Trained regions of the film will light up more brightly than untrained regions. (e) As the test image of the digit 7 will hit exactly the trained regions, the total output will be the strongest. (f) The total output signal of the collective of nanocrystals for each of the test images. The nanocrystal film **remembers** the image with which it was trained, and **calculates** the match with the test images. This simple neuromorphic procedure is three orders of magnitude more power-efficient than digital imaging.

Recommended modules: Optics, Data processing

Marijn van Huis

Project 34: Elektronentomografie van nanodeeltjes

Onderzoeksgroep: Soft Condensed Matter and Biophysics (SCMB)

Contactpersonen: Prof. dr. Marijn van Huis, Dr. Savannah Turner. Email: M.A.vanHuis@uu.nl

Tomografie is een methode waarbij uit vele tweedimensionale projecties van een object, een driedimensionaal (3D) beeld van het object wordt gereconstrueerd. Denk bijvoorbeeld aan de CAT scan of de MRI scan in het ziekenhuis. Op de nanoschaal kunnen ook 3D reconstructies worden gemaakt, waarbij een houder met een nanodeeltje in de elektronenmicroscop wordt geplaatst, en de houder wordt gedraaid van bijvoorbeeld -60° tot $+60^\circ$ in stappen van 2° . Na iedere rotatiestap wordt met de elektronenbundel een afbeelding gemaakt, waarmee een zogenaamde ‘tilt series’ wordt verkregen. Deze tilt serie dient vervolgens als input voor tomografie-software, waarmee het 3D object wordt gereconstrueerd. Hiermee kunnen vreemde vormen van nanodeeltjes, of bijvoorbeeld open volumes binnen een nanomateriaal (porositeit), veel beter worden bestudeerd. De studenten krijgen een intro in het gebruik van de elektronenmicroscop, zullen het opnemen van een experimentele tiltserie bijwonen, en gaan verder aan de slag met de tomografie-software (tutorial met voorbeeld-tiltserie, en toepassen op een tiltserie die is opgenomen op het EM-centrum van de UU). **Variaties experiment:** Er zijn allerlei variaties op de tomografie denkbaar, zo kan bijvoorbeeld een deel van de tilt serie worden weggegooid en worden nagegaan wat het effect is op de reconstructie. Ook de reconstructie-algorithmes kunnen worden gevarieerd (type, aantal iteraties), en er kunnen allerlei keuzes worden gemaakt bij de 3D visualisatie van de reconstructie (thresholding, isosurfaces).

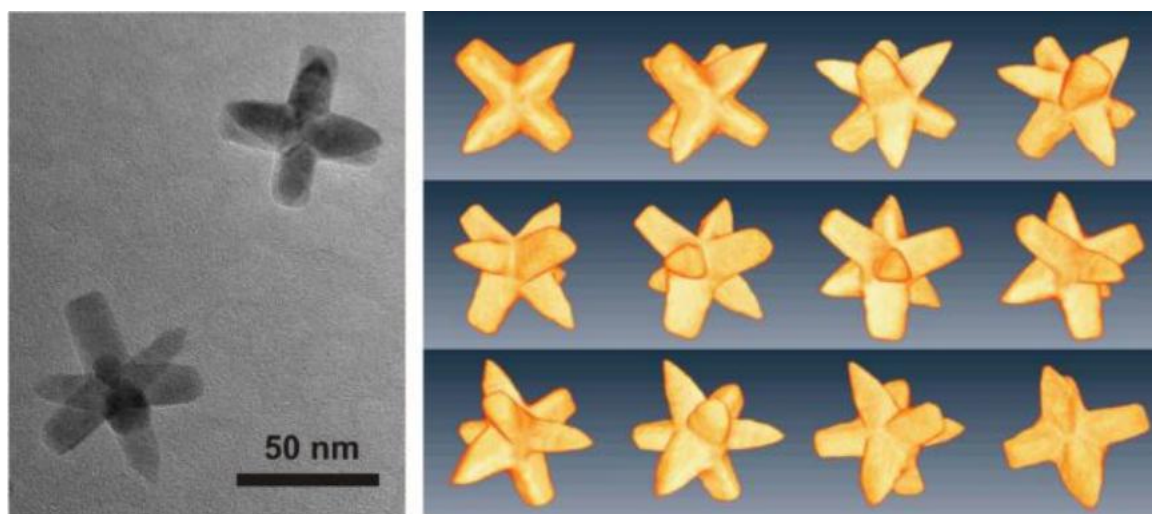


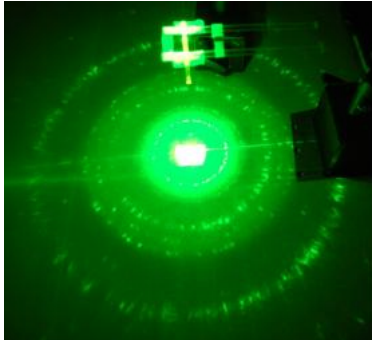
Figure. Octapods of approximately 60 nm in size, consisting of a CdSe core and eight CdS branches (pods). Left: EM image showing a projection of two octapods. The morphology is not obvious. Right: Perspective views of a tomographic 3D reconstruction of a single octapod; the structure can be viewed from any angle and the shape can be analyzed.

Recommended modules: Data processing, optics, mass spectrometry

Arnout Imhof

Contact: Dr. Arnout Imhof, a.imhof@uu.nl

Project 35. Soft crystallography with light



Colloidal particles have sizes from nanometers to micrometers. They can spontaneously form crystalline materials when allowed to come to equilibrium. They produce beautiful colors that result from diffraction of visible light, just like natural opals. In this project you can:

- develop a setup to determine their crystal structure and lattice spacing using laser light
- measure the size of the particles and relate this to the crystal
- measure the compressibility of these soft crystals

Recommended modules: Optics, Data processing

Project 36. When light gets drunk

Contact: Dr. Arnout Imhof, a.imhof@uu.nl

We see most objects because they modify the direction and the spectrum of light by scattering and absorption. That is why paper and milk look white, but you hit your head against a glass ceiling. And when you cycle in the rain your jeans will look a deeper shade of blue. Inside most materials light follows a complicated path similar to a drunkard's walk. In this project you can:

- measure the spectrum of light transmitted and reflected by objects of your choice
- measure the angular dependence of light emerging from the objects
- try to figure out how the drunken light walked



Recommended modules: Optics, Data processing

Arnout Imhof & Alfons van Blaaderen

Contact: Dr. Arnout Imhof, a.imhof@uu.nl, Prof. dr. Alfons van Blaaderen, a.vanblaaderen@uu.nl

Project 37. Efficiency of random closest particle packings



Particle packings play an important role in physics, chemistry and materials science and technology. In areas from the transportation of bulk goods and the densification of ceramic materials to the formation of glassy or amorphous solids, we want to know how many particles can fit maximally in a given volume. Especially for particle packings where even the local order is unlike that of crystals, also referred to as random (close) packings. In this project we challenge you to explore the packing efficiency of granular particles of various regular ‘model’ shapes such as marbles (spheres), orzo pasta (prolate ellipsoids), and M&Ms (oblate ellipsoids). The packing density depends on factors like particle monodispersity, shape, container geometry, and packing equilibration by tapping. Can you find a phenomenological description of particle packing densities of these model shapes? If time allows, these granular model systems can *also* be used to determine 3D regular or crystalline particle packings, can you design an experimental procedure to also study these?

Recommended modules: Optics, Data processing

Alfons van Blaaderen

Onderzoeksgroep: Soft Condensed Matter & Biophysics

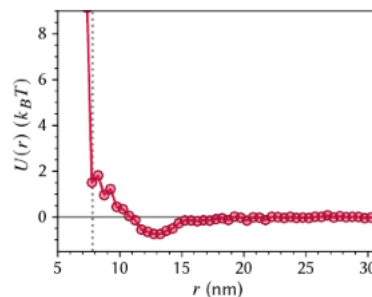
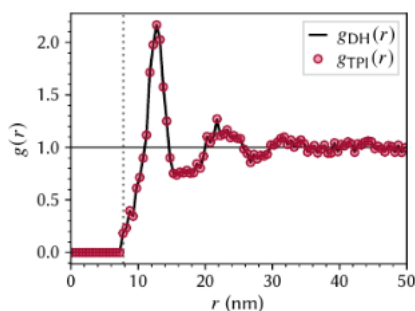
Contact: Prof. dr. Alfons van Blaaderen, Email: a.vanblaaderen@uu.nl

Project 38: Measuring pair potentials between colloidal particles

Already for over 100 years when Perrin used colloidal model particles to determine Avogadro's constant using equations and predictions derived and made by Einstein, it is clear that such particles can provide a powerful model system in condensed matter physics.

In this project we want to explore not the single particle properties as was done the beginning of the previous century, but instead investigate using microscopy and gravity 2D model systems in which particles interact, if we can experimentally measure interparticle pair potentials. We will (try to) determine and learn:

- Why is the phase behavior for 2D and 3D systems fundamentally different?
- How does this affect the interaction potentials between colloidal particles?
- Can we measure pair potentials using image processing and statistical mechanics experimentally in dilute 2D particle systems?
- How can computer simulations and theory be used in our study.
- How could we measure pair potentials experimentally in concentrated 2D particle systems?
- How can we connect this research to another project in which we try to experimentally the 2D phase behavior of colloidal particles?



Figures from: DOI: [10.33540/2252](https://doi.org/10.33540/2252)

Recommended modules: Optics, Data processing

Alfons van Blaaderen

Onderzoeksgroep: Soft Condensed Matter & Biophysics

Contact: Prof. dr. Alfons van Blaaderen, Email: a.vanblaaderen@uu.nl

Project 39: Phase behavior of a 2D colloidal model system

Already for over 100 years when Perrin used colloidal model particles to determine Avogadro's constant using equations and predictions derived and made by Einstein, it is clear that such particles can provide a powerful model system in condensed matter physics.

In this project we want to explore not the single particle properties as was done the beginning of the previous century, but instead investigate using microscopy and gravity the phase behavior of a system of strongly interacting 2D colloidal particles. We will (try to) determine and learn:

- Why is the phase behavior for 2D and 3D systems fundamentally different?
- How can we use light microscopy and image processing and to determine local properties of interacting particles experimentally.
- Study the phase behavior of a 2D model system.
- How can computer simulations and theory be used in our study.
- Can we extend the model system to also investigate 2D quasicrystals?

How can we connect this research to another project in which we try to experimentally measure pair potentials between colloidal particles?

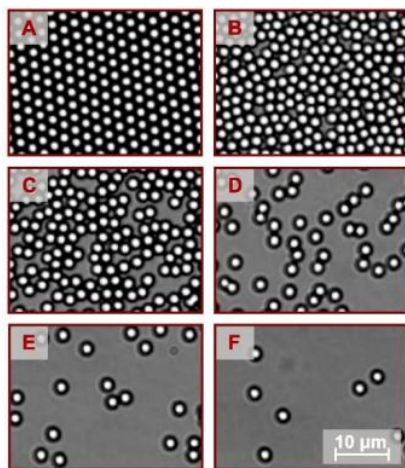


Image taken from: DOI: [10.1016/j.jcis.2021.09.116](https://doi.org/10.1016/j.jcis.2021.09.116)

Recommended modules: Optics, Data processing

Gerhard Blab

Contact: Dr. Gerhard Blab, email: G.A.Blab@uu.nl

Projects 40-41. OCT Data Analysis Project for the Bachelor Course “Experiment Design”

Optical Coherence Tomography (OCT) is a powerful microscopy method that uses interference and spectroscopic analysis to produce high resolution depth information of a sample. For this purpose, it is often used in ophthalmology, the medical field dealing with disorders of the eye, to diagnose diseases in the retina (the light-sensitive layer at the back of your eye) or the optic nerve. The OCT we use in Utrecht (see figure 1a) is of the spectral domain type and consists of a confocal microscope with a broadband ($\lambda_0 = 840 \text{ nm}$, $\Delta\lambda = 160 \text{ nm}$) light source for illumination and a very fast fiber-coupled spectrometer to collect the interference spectra returned from the sample. The depth information is then obtained by Fourier-transforming these spectra (figure 1e and f). During her thesis research, PhD candidate Pegah Asgari (now dr. Asgari) pushed the resolution of the limit: by heating a $10 \mu\text{m}$ polystyrene beads with a second laser, she caused the beads to thermally expand. Using our OCT (to be exact: the phase of the complex Fourier Transform of the spectra obtained), she could follow the expansion and contraction of the beads on a scale of a nanometer or less! Her results are shown in figure 2.

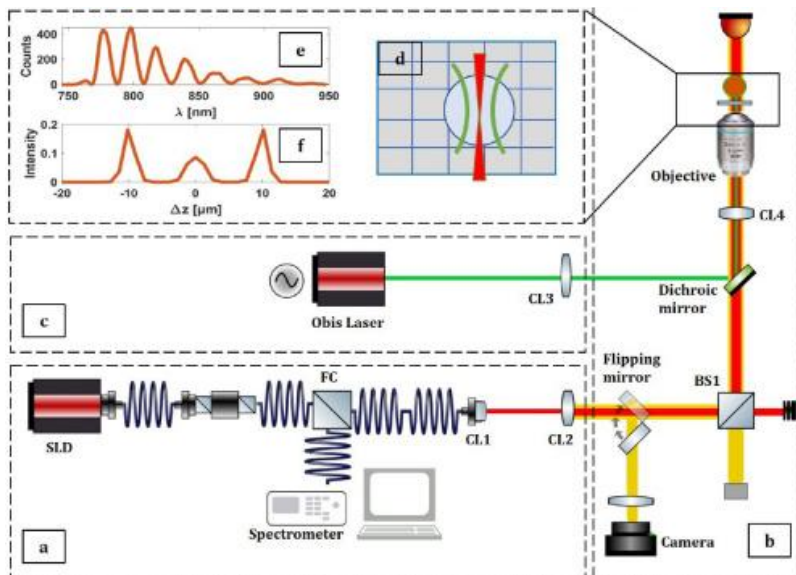


Fig. 1. Setup of the SD-OCT system. (a) The light source and detection part that includes a superluminescent light source (SLD), a fiber coupler (FC) (two practical options 90:10 or 50:50), a collimator (CL1), and spectrometer (SP). (b) Microscopy section where the light is directed to the sample by a collimated lens (CL2), BS1 (P), microscope objective, and arrives at the microsphere. (c) excitation arm with a 532-nm laser, with collimation lens (CL3). (d) Schematic of the sample (e)

Measured interferogram formed by the top and bottom reflections from the microsphere and (f) its intensity Fourier transform

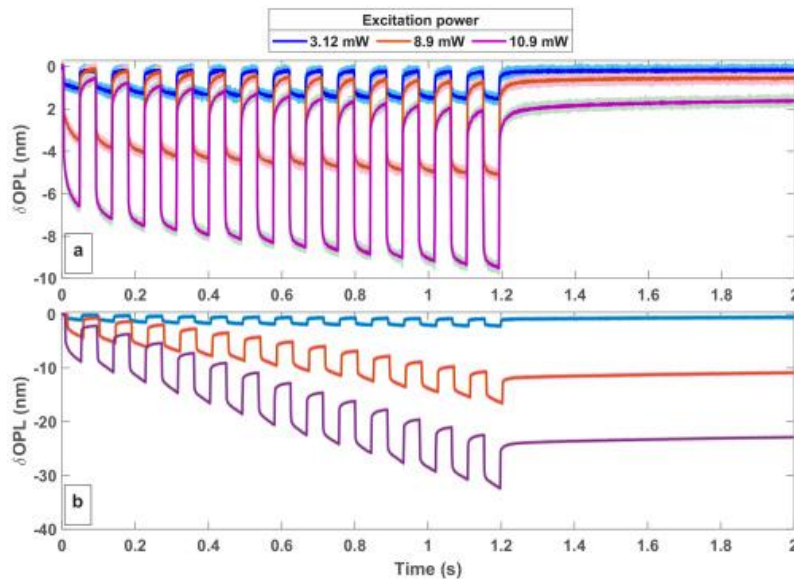


Fig. 2. Optical path length of two microspheres exposed to various power, as example of two different microspheres behavior (a) Microsphere showing a low amount of inelastic creep. (b) Microsphere showing significant creep. Note the different scale of the y-axis. Raw data is shown in lighter color behind the smoothed data (moving average 50 data points)

Both figures from: Asgari et al. “Sub-nanometre measurement of transient structural changes in dye-doped polystyrene microspheres”, *Optics Continuum*, Vol 2, Issue 2, pp 259–265 (2023); DOI: 10.1364/OPTCON.47770

Disclaimer: Both projects offered deal with the analysis of OCT data – due to the renovation in the Ornstein Laboratory, it is currently not possible for students to enter the labs and do measurements by themselves. You will therefore mostly work with ‘synthetic’ data, i.e. simulated ideal data; though if you progress fast enough, we might be able to apply your knowledge to some already measured experimental data! As mentioned before, an important step in the analysis of OCT data is the Fourier transformation. In most OCT papers – such as the one by Pegah – you will find a mathematically elegant and exact derivation of why this works. However, for a real experiment we cannot use the ideal Fourier transform (we do not have an infinite amount of data ...) but only the discrete (limited # of data) ‘Fast Fourier Transform’ or FFT, which leads to some issues. For both projects, some preliminary work has been done by last year’s students of what was then called the “Onderzoeksstage”, so you will have a report and some python code from which to start.

Requirements:

- read the paper by Asgari et al. on OCT (DOI: 10.1364/OPTCON.477703)

- read up on the (Fast) Fourier Transform
- know how to program in Python (some code already exists and can be adapted by you!)

Project 40 – Where is the maximum of an FFT? If you choose the first project, your task will be to quantify the size of a bead in OCT data by finding the position of the maximum of the FFT. The group last year identified several possible algorithms that work for this purpose.

Your task will be to test these algorithms and find out whether these algorithms still work well

- if the synthetic data is less idealized and resembles real data more closely; and
- if there is noise of different types in the synthetic data.

Recommended modules: Optics, Data processing

Project 41– What exactly is ‘size’ in OCT? As OCT uses interference, it technically does not report on absolute sizes of objects but rather the Optical Path Length (OPL). The OPL is the product of the optical (or refractive) index n and the physical distance d . If we want to know this distance d (which in this case is the size of the bead), we need to know the optical index n . Luckily, OCT gives us – at least in principle – a way to measure the optical index: the Fresnel equation connect the intensity of the reflected light with the optical index of the bead (more accurately: difference in index of the bead and that of the medium)!

If you choose the second project, your task will be to:

- find a reliable algorithm that extracts the correct amplitude from an FFT peak (spoiler: this is not the same as the maximum value!). This may require some literature study.
- evaluate the reliability of the found algorithm (error as function of OPL, ...)
- use the Fresnel equations to calculate the optical index from the found amplitude of the FFT; evaluate what is the smallest change in optical index that could be measured this way

Recommended modules: Optics, Data processing

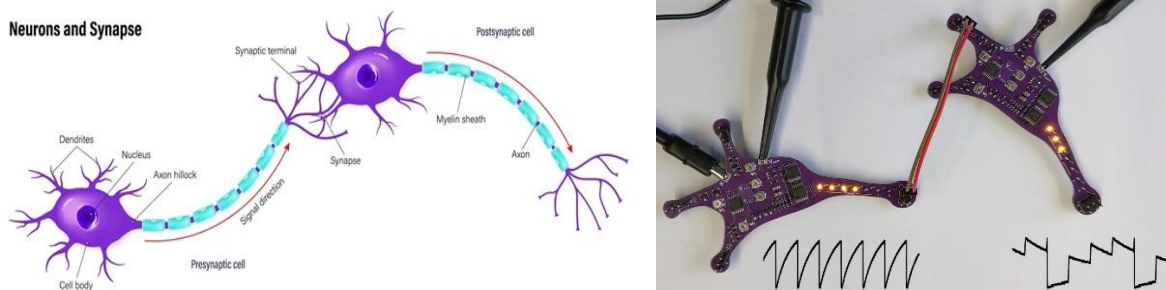
Marjolein Dijkstra

Onderzoeksgroep: Soft Condensed Matter and Biophysics (SCMB)

Contactpersonen: Prof. dr. ir. Marjolein Dijkstra, Email: m.dijkstra@uu.nl

Projecten 42 en 43: Bouwen van neurale netwerken met elektrische neuronen

Ons menselijk brein is een neurale netwerk, waarin neuronen de essentiële elementen zijn. Door de verbinding van neuronen in een netwerk kan informatie worden opgeslagen, kunnen berekeningen worden gedaan, en uiteindelijk zelfs ideeën worden gevormd. Een van de aantrekkelijke kanten van biologische neurale netwerken is dat ze veel energie-zuiniger zijn dan bijvoorbeeld computers die met chips (gebaseerd op elektrische transistoren) werken. Over het functioneren van neurale netwerken is nog heel veel te leren: hoe werken de verbindingen (synaptische functie), de activering, communicatie, etc. Het is mogelijk om de werking van biologische neurale netwerken te simuleren met zogenaamde elektrische neuronen. De opdracht hier is om van elektrische neuronen een netwerk te bouwen, en te proberen om biologische eigenschappen van neuronen na te bootsen zoals synchronisatie, synaptische plasticiteit, coincidence detectie, en leaky integrate and fire, en om zo basiseigenschappen van het netwerk te genereren, zoals een leerfunctie, geheugenfunctie, en/of het netwerk te trainen om te reageren op een bepaalde input of een machine-learning taak zoals classificatie of regressie. De onderzoeksvraag kan deels door jullie zelf worden opgesteld.



Project 42: Onderzoeksvraag 1 (TBA)

Project 43: Onderzoeksvraag 2 (TBA)

Recommended modules: Arduino, Elektronica, Data-verwerking

6. Group Project Proposal Handout

Purpose. This handout supports the ERP group proposal due on 15 May 2026 . It combines a student-facing assignment guide with a teacher/TA feedback guide, using shared structure and wording so expectations, feedback, and support are aligned.			
Deadline 15 May 2026	Length max. 3 A4 pages	Submission one proposal per group	Use this for student planning + teacher/TA feedback
Student Handout			
Aim for students. Use this handout as a guide to prepare your proposal. The proposal shows what your group wants to investigate, how you plan to do it, and how you will work together. Your proposal should be specific, realistic, and open to improvement. Please simply copy the form below and fill the point 1-6 to finish the assignment, and add your group number/student name in the end.			
1. Project title	Choose a short, informative title that signals the main topic or experiment.		
2. Research aim and question	State the main question or goal. Explain what you want to measure, test, build, compare, or optimize, and why it is scientifically interesting.		
3. Experimental approach	Explain how you plan to address the research question. Name the setup, method, instrument, or model you will use, the quantities you will measure, your first test or pilot step, and the main risks or uncertainties.		
4. Feasibility and planning	Show that the project is realistic within the available time, equipment, expertise, and supervision. Include the first steps and a short timeline for the next phase.		
5. Collaboration plan	Describe how your group will collaborate using four focus points: Task, Working Method, Working Atmosphere, and Coaching.		
6. Reflection	Conclude with what is already strong in your plan, what remains uncertain, and what you expect to improve in the next phase.		

Collaboration plan: four focus points

Task	Working Method	Working Atmosphere	Coaching
What do we have to deliver?	How do we divide tasks?	How do we communicate respectfully?	What do we expect from teacher and TA?
What is our shared goal?	How and when do we meet?	How do we give feedback?	When and how do we ask for help?
What is each person's contribution?	How do we document progress and data?	How do we handle disagreement or imbalance?	How do we prepare for meetings and use feedback?

See what you concluded from the course below:

Criteria for effective collaboration			
<u>Task</u>	<u>method</u>	<u>atmosphere</u>	<u>coaching</u>
- Take the time - Communicate - Read the assignment - GOOD INVESTIGATION OF THE PROBLEM - Plan beforehand/sketch outline	- Compromise - Equality - Sketch situation - democratic - Being Decisive - Present reflection - Leadership - getting on the same page	- Respect each other - Stay calm - Clear expectations from each other - face to face - Humour and clarity in communication - friendly banter	- nice - Able to ask questions - Positivity - not being made fun of by individuals (thanks Peter :))

What makes a strong proposal? A strong proposal has a clear and feasible research goal, a well-connected method, thoughtful planning, clear agreements about collaboration, and awareness of documentation, uncertainty, and improvement.

7. Rubric Experimental Research Project 2025 - 2026

Version 10-04-2026

Score	Insufficient	Sufficient/good	Excellent
Criteria			
Modules			
1. The student follows and completes one of the elective modules on experimental techniques and methods - follows and completes one of the elective modules on Electronics and signal processing; Laser and optics; Data processing; Mass spectroscopy; Proto lab; Arduino - familiarises herself/himself with the instrumentation used in the module and understands the functionality and caveats of the methods and techniques treated in the modules (module-dependent)	- Data Acquisition: setups are incomplete or flawed - Signal Processing & Statistical Analysis: analysis is incomplete or flawed - Spectroscopy & Microscopy: poor understanding of methods	- Data Acquisition: setups are adequate - Signal Processing & Statistical Analysis: processing and analysis of signals are correct - Spectroscopy & Microscopy: Successful measurement and analysis	- Data Acquisition: flawless and proficient construction of setups - Signal Processing & Statistical Analysis: Superior understanding and application of theory - Spectroscopy & Microscopy: High-quality recording and processing of data
Research Project			
2. The student defines and formulates an experimental project - formulates an empirically verifiable research question or goal that can be answered or met within the boundary conditions of the project (available time, budget, equipment, expertise) - designs an instrument or setup for an experimental research project given the conditions of the project - searches and reads literature during the definition of the research question or goal and during the design of the experiment	- Research goal not formulated, not verifiable or unrealistic given boundary conditions - Proposed method and setup are incomplete, inadequate or unsupported - No clear picture of relevant quantities and how they connect to measurements to be performed - No or improper literature was used in the preparation phase - Probability of successful execution is low	- Research goal is clear and realistic given boundary conditions - Proposed method is adequate and backed by documentation supporting the design of the setup and experiment and good estimates - Relevant quantities are defined and connected to the experiment - Proper literature was found and studied	- Research goal is clear, realistic and innovative - Thought-through experimental method, documentation and discussion demonstrate good understanding of the proposed setup and experiment - Relevant quantities are well defined and connected to the experiment - Key literature was found, studied and implemented in the project proposal
3. The student plans and executes an experimental project in collaboration - plans and controls a complex experimental project from formulation through construction, test and execution all the way to completion and reporting - learns to deal with the group dynamics that comes with an (experimental) research project (e.g. division of tasks and responsibilities, coordination, exchange of ideas and information, accountability)	- Messy or incorrect execution of the experimental project - Problems met during the execution of the project are not resolved or only with excessive help - No inventivity - Project suffers from bad planning and long delays - Poor group work with no or limited mutual interactions	- Successful execution of the experimental project - Able to resolve problems met during the execution of the project, with appropriate amount of help - Project is properly planned and runs on time - Good group work with regular mutual interactions	- High-level and effective execution of the experimental project - Able to resolve problems met during the execution of the project, with no or limited help - Project is planned excellently so that an extraordinary amount of work is done - Effective and synergetic collaborative spirit leading to proficient results
4. The student masters instrumentation - chooses the proper instruments and procedures to achieve the desired accuracy and precision - calibrates instruments throughout the project - understands the physical working principle of the (commercial) instruments, sources and / or detectors used and the possible systematic effects that these introduce	- Improper handling of equipment - Missing or incorrect calibrations or preparation procedures - Poor understanding of how and what instruments are actually measuring	- Proper handling of equipment - Proper calibrations and preparation procedures - Good physical understanding of what the instruments are recording and in what range	- Flawless handling of equipment - Student has a role in selection, construction and/or testing of instrumentation - Proper calibrations and preparation procedures

	- No idea of limitations, precision or possible systematic errors of instrumentation	- Basic idea of limitations to the instrumentation	- In-depth physical understanding of the physical functioning of the instruments - Good insight into limitations and relevant systematic errors
5. The student is able to construct an experimental setup* - builds hand-on experience on building setups and using measurement instruments - builds experience on resolving issues with the setup - is aware that a constructed setup has its limitations and possible systematic errors	- Incorrect building or assembly of the experimental setup - Unable to troubleshoot, change or redesign the experimental setup to meet the research goals - Test measurements of poor quality, not demonstrating mastery of the setup and giving no information for choices in execution - No notion of the possible limitations of the setup and sources of uncertainty and error	- Building or assembly of a properly functioning experimental setup providing sufficient results - Showing the ability to troubleshoot, change or redesign the experimental setup - Proper test measurements demonstrating control of the setup and giving indications for good choices in execution - Basic (qualitative) notion of the (im)possibilities of the setup and sources of uncertainty and error	- Step-wise design of a proficiently functional experimental setup relevant to answering the research question - Good evolution of the design process - High-quality test measurements demonstrating mastery of the setup and strongly supporting choices in execution - Awareness of the flaws and benefits of the experimental setup - Sources of uncertainty and systematic errors are quantified
6. The student takes part in peer consultation and discussion - studies the work of peers substantively in different phases of the project and delivers constructive critique - discusses the research project with peers and is capable of deciding which suggestions are improvements and which are not	- No or only negative feedback given to peers - Criticism that is not to the point or not on the content of the project	- Good and relevant discussion with peers about the downsides and benefits of the experimental project - Interested and participative attitude in providing feedback	- Good and fruitful discussion with peers about the downsides and benefits of the experimental project - Inspiring and participative positive attitude in providing feedback
7. The student demonstrates conceptual understanding - is capable of developing a theoretical / computational model for the case that is studied experimentally - can estimate prior to execution of an accurate experiment what the outcomes will be, through calculations and rough estimates given the input values for certain physical quantities - works synthetically and applies (theoretical, technical or conceptual) knowledge from modules and physics in general in the design of experiments and in the analysis and interpretation of the results - formulates scientific arguments following from the research question or goal, the collected and analysed evidence and conclusions based on results	- Physical model is not presented or irrelevant to the experiment - Prior to accurate measurement runs modeling is too limited to yield clear expectations for outcomes - Student relies heavily on TA or instructor for physical interpretation of data and observations - Unable to formulate conclusions based on data and observations - Argumentations are weak	- Physical model is correct and quantities are defined and connected to the experiment - Prior to accurate measurement runs modeling is sufficiently developed to yield clear expectations for outcomes - Student applies own knowledge for physical interpretation of data and observations - Can formulate conclusions based on data and observations - Argumentations are strong	- Physical model is correct and quantities are defined and connected to the experiment - Prior to accurate measurement runs modeling is developed such that accurate expectations for outcomes are available - Student applies own knowledge for physical interpretation of data and observations and reaches excellent conceptual understanding - Conclusions follow logically from physical interpretation of data and observations - Solid argumentation
Reporting and communication			
8. The student communicates regularly, effectively, and respectfully, and reports results accurately, convincingly, and scientifically sound** - convincingly discusses conclusions and argumentations (founded on theoretical background, experimental research and logical thinking) orally with the supervisor - has an authentic research experience and experiences methods and activities common in research groups - communicates regularly, effectively, and respectfully with (technical) staff	- No or very little communication and interaction with the supervisor, teaching assistant and/or technicians - Incomplete or inadequate reporting on progress and outcomes - Discussions are largely procedural and students rely on insights and interpretations of supervisor	- Regular and adequate communication and interaction with the supervisor, teaching assistant and/or technicians - Complete and adequate reporting on the progress and outcomes - Discussions are largely scientific and students demonstrate own insights and interpretations	- Efficient and fruitful communication and interaction with the supervisor, teaching assistant and/or technicians - Efficient, complete, and useful reporting on progress and outcomes - Discussions are scientific and profound

- reports in a scientifically sound way: does not communicate misleading results and does not commit plagiarism	- Inappropriate or non-respectful tone in communication	- Respectful in communication	- Interested attitude and respectful in communication
9. The student documents the project in collaboration at Bachelor Research project level - Documentation includes at the least a (collective) lab journal and a file structure in which all the collected information and data is organized clearly, and furthermore e.g. notes, memos, reports and presentations for the project group	- From documentation, it is not possible to track development of the project or reproduce findings - Digital document structure is absent - Crucial information is not recorded - Evidence for validity of results and protocols is missing	- It is possible to track development of the project or reproduce findings on a global level - Digital document structure is present - Direct observations, analyses and interpretations are recorded - Documentation provides evidence for validity of results and protocols	- Documentation during full project “by the book”; it is possible to track development of the project or reproduce findings on a detailed level - Digital document structure that is also functional to externals - Besides direct observations, analyses and interpretations, also synthesis of information and incremental planning and ideation are properly recorded - Evidence for validity of results and protocols inherent to documentation
10. The student creates and presents a poster - can present a poster to fellow researchers - uses proper technical terminology, results including uncertainties, tables and figures and schematics of the setup to make research accessible to peers	- Uninterested presentation of the poster - Poster is unclear or incomplete in content - Poster is unappealing to look at - Fonts, graphs or figures unreadable from some distance	- Adequate presentation of the poster - Good answering of questions - Poster provides a good overview of the research project, including method, setup, and outcomes - Poster is inviting and can be read from some distance	- Enthusiastic presentation of the poster - Good and to-the-point answering of questions - Poster provides a good and complete insight into the research project, including method, setup, and outcomes - Poster is esthetically appealing and can be read from some distance
11. The student reflects on own work and achievements during the project and afterwards - monitors the quality of results and improves the design of the experiment during the project - judges and improves own work based on the supplied rubrics and demands - identifies possible, interesting and relevant follow-up research that follows logically from the conducted research (possibly requiring new equipment or other conditions)	- No (self)reflection shown during nor after the execution of the project - Monitoring and improvement depends solely on TA and supervisor input - No or unrealistic suggestions for follow-up research	- The project is well monitored, students are aware of quality and progress - Points for improvement are formulated and regularly implemented in the experiment - Realistic suggestions for follow-up research are made	- The project is well monitored with a critical attitude - Good reflection on progress as well as self-reflection on the student’s own commitment - Points for improvement are identified and promptly implemented to improve experimentation - Genuinely interesting suggestions for follow-up research are made

* Creating a physically safe workspace for yourself and others is a prerequisite for working in the lab. The mandatory Science lab safety instructions can be found in the ULearning environment (login with solis-id):

- <https://ulearning.uu.nl/course/view.php?id=892> (Dutch)
- <https://ulearning.uu.nl/course/view.php?id=976> (English)

Besides these general guidelines, specific rules can apply in several projects. Lab staff and TA’s monitor the behaviour in the lab room. Repeatedly failing to comply with regulations and directives or causing hazardous work circumstances leads to termination of the project.

** In cases of plagiarism (copying work from fellow students or literature while presenting it as own work) or fraud (data fabrication or manipulation), the course coordinator will inform the Board of Examiners (BoE) in writing. The BoE then decides on whether fraud or plagiarism was committed, and which measures are required.

More information can be found in the Onderwijs- en Examenregeling (OER), Art. 5.14

<https://students.uu.nl/sites/default/files/Onderwijs%20en%20Examenregeling%20Undergraduate%20School%20of%20Science%202024-2025.pdf>